

COMENIUS UNIVERSITY IN BRATISLAVA
FACULTY OF MATHEMATICS, PHYSICS AND INFORMATICS

MACHINE LEARNING ASSESSMENT OF THE
TRADING VALUE OF CENTRAL BANK
INFORMATION
BACHELOR THESIS

2026

JAKUB NOVOTNÝ

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Study Programme: Data Science
Field of Study: Computer Science and Mathematics
Department: Department of Computer Science
Supervisor: Ing. Pavel Gertler, PhD.

Bratislava, 2026
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Univerzita Komenského v Bratislave
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Študijné odbory: informatika
matematika
Typ záverečnej práce: bakalárska
Jazyk záverečnej práce: anglický
Sekundárny jazyk: slovenský

Názov: Machine Learning Assessment of the Trading Value of Central Bank Information
Využitie modelov strojového učenia pri oceňovaní hodnoty informácií centrálnej banky

Anotácia: Bakalárska práca sa bude venovať tomu, ako finančné trhy absorbujú a oceňujú informácie z oficiálnej komunikácie centrálnych bánk. Na textové dátové súbory budeme aplikovať techniky spracovania prirodzeného jazyka (NLP) a modely strojového učenia, pričom sa zameriame na tlačové konferencie ECB. Empirická časť extrahuje z týchto textov signály z oblasti menovej politiky a konštruuje sentiment. Otestujeme rôzne prístupy k predspracovaniu dát, tematickej klasifikácii a kontextovej analýze. Práca bude následne sledovať, ako sa extrahované signály vyvíjajú v čase a ako interagujú s kľúčovými makroekonomickými a trhovými premennými, ako aj s mierami trhovej neistoty. Vyhodnotíme, do akej miery rôzne formáty a fázy komunikácie predikujú udalosti na trhu. Posúdime tiež silné stránky a obmedzenia rôznych architektúr strojového učenia pri analýze finančných a ekonomických textov.

Okrem akademického prínosu vytvorí táto práca základ pre jeden z výskumných projektov v NBS zameraného na informačnú hodnotu komunikácie centrálnych bánk. Úspešné výsledky môžu priamo prispieť k výskumu relevantnému pre menovú politiku a potenciálne viesť k spoluautorskej publikácii s NBS..

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THESIS ASSIGNMENT

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Title: Machine Learning Assessment of the Trading Value of Central Bank Information

Annotation: The thesis will examine how financial markets absorb and price information from official central bank communications. It will apply Natural Language Processing techniques and machine learning models to textual datasets, focusing on ECB press conferences. The empirical work will extract monetary policy signals and sentiment from these texts. The goal is to test different approaches to data preprocessing, topic classification, and contextual analysis. The thesis will trace how extracted signals evolve over time and interact with key macroeconomic and market variables and measures of market uncertainty. It will evaluate how well different communication formats and phases predict market events and responses. It will also assess the strengths and limitations of various machine learning architectures for financial and economic text analysis.

Beyond its academic contribution, the thesis will lay the groundwork for an NBS research project on the informational value of central bank communications. Successful results may feed directly into policy-relevant research and potentially lead to a co-authored publication with NBS.

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Declaration: I hereby declare that I have independently written this entire bachelor thesis entitled “Machine Learning Assessment of the Trading Value of Central Bank Information”, including all its figures and appendices, using the literature listed in the attached bibliography.

In preparation of the thesis I have also used the following artificial intelligence tools: Gemini and Claude for the better verbal expression, helping with source searching and writing the Python code for making the plots. I have used artificial intelligence tools in accordance with relevant legal regulations, academic rights and freedoms, ethical and moral principles while simultaneously maintaining academic integrity. I am aware that I am fully responsible for the accuracy of the final text.

Acknowledgments: I would like to thank to my supervisor Ing. Pavel Gertler, PhD, for his advice, spent time and interesting conversations about macroeconomy. Last but not least, I would like to thank my family, friends and girlfriend for enduring my emotional outbursts, encouraging me and helping me finish my bachelor’s degree.

Abstrakt

Centrálne banky, ako je Európska centrálna banka, neurčujú iba úrokové sadzby — používajú aj jazyk na to, aby usmerňovali očakávania finančných trhov o budúcnosti. Táto práca skúma, či slová samotnej Európskej centrálnej banky dokážu predpovedať vývoj úrokových sadzieb a reakcie trhov. Pomocou metód spracovania prirodzeného jazyka sme prečítali všetky tlačové konferencie Európskej centrálnej banky z rokov 1998 až 2024 a emocionálny tón každej pasáže sme zmerali dvoma špecializovanými jazykovými modelmi. Metodologickou inováciou je rozdelenie každej tlačovej konferencie na dve časti: vopred pripravený prejav, ktorý predstavitelia banky starostlivo nacvičia, a spontánny rozhovor, v ktorom novinári kladú nepripravené otázky. Zistili sme, že novinári sú systematicky pesimistickejší ako samotní centrálni bankári a že ignorovanie tohto rozdielu skresľuje signál o menovej politike v starších štúdiách, ktoré obe časti spájajú do jedného dokumentu. Keď oba hlasy oddelíme, emocionálny tón centrálnej banky funguje ako systém včasného varovania: modely strojového učenia natrénované na tomto textovom signáli dokážu niekoľko mesiacov vopred predpovedať, či nasledujúce rozhodnutie o úrokovej sadzbe bude zvýšenie, zníženie alebo ponechanie bez zmeny. Na vzorke 289 tlačových konferencií stúpila krížovo validovaná metrika AUC pri predpovedaní smerovania ďalšieho rozhodnutia ECB o úrokových sadzbách z 0,682 (iba na základe histórie sadzieb) na 0,763 (model náhodného lesa využívajúci príznaky sentimentu z modelov FinBERT a CentralBankRoBERTa).

Kľúčové slová: analýza sentimentu, spracovanie prirodzeného jazyka, menová politika, prediktívna analýza, komunikácia centrálnej banky

Abstract

Modern central banks do more than set interest rates — they also use language to shape what financial markets expect about the future. This thesis asks whether the words spoken by the European Central Bank can predict where interest rates and market expectations will move. Using natural language processing, we read every European Central Bank press conference from 1998 to 2024 and measured the emotional tone of each passage with two specialised language models. The methodological innovation is to split each press conference into two parts: the prepared speech, which bank officials write and rehearse in advance, and the spontaneous interview that follows it, in which journalists ask unscripted questions. We find that journalists are persistently more pessimistic than the central bankers themselves, and that ignoring this difference contaminates the policy signal in earlier studies that treat the two as a single document. Once the two voices are separated, the emotional tone of the central bank acts as an early-warning system: machine-learning models trained on this textual signal can predict, several months in advance, whether the next interest-rate decision will be a hike, a cut, or no change. Across 289 press conferences, the cross-validated AUC for predicting the direction of the next ECB rate decision rises from 0.682 (rate history alone) to 0.763 (random forest with FinBERT and CentralBankRoBERTa sentiment features).

Keywords: sentiment analysis, natural language processing, monetary policy, predictive analysis, central bank communication

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List of Abbreviations

AUC Area Under the Curve.

ECB European Central Bank.

FED Federal Reserve.

GC Governing Council.

GFC Great Financial Crisis.

IS Introductory Statement.

MP Monetary Policy.

MPS Monetary Policy Statement.

MRO Main Refinancing Operations.

NLI Natural Language Inference.

NLP Natural Language Processing.

OIS Overnight Indexed Swaps.

Q&A Question and Answers.

QE Quantitative Easing.

ZLB Zero Lower Bound.

Introduction

The modern practice of monetary policy is, to a large extent, an exercise in communication. Since the abandonment of strategic opacity in the 1990s [28], central banks have come to treat the management of expectations as a policy instrument in its own right [4, 6]. Each scheduled press conference of the European Central Bank (ECB) is therefore a market-moving event: the textual record is not a description of past decisions but the channel through which the central bank shapes expected future rates, anchors inflation expectations, and influences financial conditions at horizons beyond its direct operational reach [17].

The empirical study of these communications has progressed substantially along computational lines. Early work relied on dictionary-based methods that approximated tone by counting predefined positive and negative terms; the seminal observation of Loughran and McDonald [23]—that words such as *liability* carry no negative connotation in financial filings—made clear that general-purpose lexicons systematically misclassify domain-specific discourse. Transformer-based language models have since superseded this paradigm, evaluating each token in relation to its neighbours and resolving negation and qualification that bag-of-words methods cannot capture. In the central-bank domain, FinBERT [2]—a BERT-base encoder fine-tuned on the Financial PhraseBank—and CentralBankRoBERTa [25]—a RoBERTa-base encoder fine-tuned on central-bank speeches—now define the operational standard for sentiment quantification, while recent stance classifiers [26, 27] and the BIS central-bank language models [12] extend the frontier toward hawkish/dovish labels.

What the literature still routinely treats as a single object, however, is the press conference taken as a whole. Yet an ECB press conference is in fact two structurally different communicative acts: a prepared Introductory Statement (IS), read by the President and exhaustively pre-agreed by the Governing Council, and an unscripted Question and Answers (Q&A) session in which journalists set the agenda [24]. Treating the two as one document mixes the institution’s official line with the structural pessimism of the press corps and obscures which phase carries the policy signal. The contribution of this thesis is to take that distinction seriously. We build a custom pipeline that isolates the IS from the Q&A and separates journalists’ questions from the President’s answers, quantify the tone of each segment in parallel with FinBERT and CentralBankRoBERTa,

and ask whether the resulting series carries information about the ECB’s policy-rate cycle and the Overnight Indexed Swaps (OIS) curve—a market-implied term structure of expected future rates—over and above the rate history itself. Beyond its academic contribution, the work lays the empirical groundwork for a planned Národná banka Slovenska research project on the informational value of central-bank communications.

Three intuitions should be held in mind from the outset. First, the ECB sets a short-term rate, but markets price an entire term structure of expected future rates, and the latter is what communication is designed to move. Second, sentiment (positive or negative tone) is not the same as stance (hawkish or dovish policy implication); the difference matters during episodes such as the Zero Lower Bound (ZLB) period of unconventional easing, when the language of asset purchases is lexically distinct from that of conventional rate cuts. Third, the IS and the Q&A carry structurally different information and must be analysed jointly but not interchangeably. Each idea receives formal treatment in the chapter that requires it.

The thesis is organised in four chapters. Chapter 1 establishes the theoretical background, covering the evolution of central-bank communication, the trajectory of Natural Language Processing (NLP) techniques in financial text analysis, and the quantitative targets—policy rates, the Wu–Xia shadow rate, OIS rates, and structural shocks—against which the textual signal will be evaluated. Chapter 2 documents the data and methodology: the scraping pipeline that produces the IS/Q&A-split corpus of 289 press conferences from June 1998 to December 2025, the chunk-level segmentation, the multi-model sentiment extraction, the zero-shot topic taxonomy, and the chunk-to-meeting aggregation. Chapter 3 reports the empirical findings in four arcs of increasing demand on the data: classifier coherence and historical anchoring, the sectional anatomy of the press conference, sentiment around the policy-rate cycle, and the predictive value of the textual signal for the direction of the next rate decision. Chapter 4 discusses the implications for policy makers and markets, reflects on the methodological choices, acknowledges the limitations, and identifies the most natural extensions—chief among them the cross-bank replication that would underpin the planned co-authored publication with the Národná banka Slovenska.

Chapter 1

Theoretical Background

This chapter establishes the theoretical foundation for the research by connecting the macroeconomic concepts of Monetary Policy (MP) with advanced language technology. First, it explores the evolution of central bank communication, emphasising the structural nuances of the ECB’s press conferences. Second, it introduces Natural Language Processing as a methodological bridge, tracing the shift from traditional sentiment lexicons to specialised transformer models capable of decoding ECB narratives. Finally, the chapter defines the quantitative targets ranging from official policy rates to high-frequency market expectations and structural shocks, providing the framework needed to assess the real-world financial impact of central bank text.

1.1 Central Bank Communication

Over the past three decades, central banking has undergone fundamental changes in its disclosure structures. Before the 1990s, there was no public explanation of why monetary policy decisions were made. Policymakers believed that a degree of unpredictability enhanced the effectiveness of policy interventions. This period of institutional unpredictability was characterised by a marked lack of formal public guidance [28, p. 3]. Market participants repeatedly had to guess the target interest rate and the bank’s policy setting. They relied on the size, timing, and nature of the operations conducted by the Federal Reserve (FED) trading departments in the open market.

Since then, communication has evolved from a peripheral public relations function to a core instrument of MP. Central banks have increasingly recognised transparency and accountability as fundamental public service obligations [4, p. 15–18]. Beyond institutional accountability, the evolution of MP frameworks has demonstrated that transparent and credible central banks are considered important for maintaining long-term price stability [6]. This is particularly evident in the widespread adoption of

inflation targeting across advanced and emerging economies. Empirically, clear communication has been shown to enhance central bank independence, strengthen institutional credibility, and anchor inflation expectations in complex macroeconomic environments [11, p. 7].

The European Central Bank Framework Within the Eurozone, the ECB faces a communication challenge: it must address a fragmented, multi-national audience while maintaining a unified monetary stance. Since its establishment in 1999, the ECB’s communication framework has progressively evolved to anchor expectations across diverse sovereign bond markets. This institutional evolution is most evident in its structured approach to policy announcements [10].

1.1.1 Monetary Policy Statement

At the operational core of the ECB’s communication strategy is the Monetary Policy Statement (MPS), which is delivered during the press conference that directly follows the MP meetings of the Governing Council (GC). The council consists of the six members of the Executive Board (including the President and Vice-President) and the governors of the euro area national central banks, who meet to deliberate and set monetary policy parameters for the entire currency bloc. The ECB currently schedules eight MP meetings per year, operating on an approximate six-week cycle [16].

Schedule of the Governing Council Meeting Day The organisation of the announcement day is standardised, reflecting the institution’s strict commitment to predictability and the mitigation of market volatility. Starting from July 21, 2022, the ECB modified its historical publication timeline to optimise the way financial markets process complex policy signals [9]. The initial policy decision—usually a short press release detailing the exact interest rate parameters, deposit facility rates, and main structural changes to asset purchase programmes—is published at 14:15 CET. Shortly afterward, the press conference begins at 14:45 CET, during which the ECB President and Vice-President present the detailed MPS and interact directly with the financial press. In parallel with the start of the Q&A segment, the full text of the MPS is published on the ECB’s digital platforms at 15:00 CET. At 17:00, European stock markets close independently of the ECB, ending the day’s reactions to its actions, as shown in Table 1.1.

Structure of the Press Conference The press conference itself is divided into two distinct communication phases: the reading of the formally prepared IS and the subsequent unscripted Q&A session. This structural division is not merely procedural;

Table 1.1: Governing Council Meeting Day Schedule.

Event	Scheduled time (CET)
Meeting of the GC	9:00 – 12:00
MP announcement	14:15
Press conference (IS + Q&A)	14:45 – 15:15
European stock market close	17:00

it has measurable consequences for financial markets, as the two sections serve different informational functions and generate distinct asset price dynamics.

The IS is a structured, carefully drafted document that represents the GC’s consensus view [24]. It provides a linear overview of the economic assessment, inflation dynamics, and the evaluation of monetary and financial conditions. The language used in this document is deliberately precise, relying on specific syntactical formulations to signal the future policy path. This practice is commonly known as forward guidance: an explicit communication tool by which the central bank provides forward-looking indications about the future trajectory of its policy rates or asset purchases, thereby shaping market expectations beyond the current meeting [4].

In contrast, the Q&A session introduces a semi-structured interaction. Advanced textual analysis and structural topic modelling of historical transcripts demonstrate that the thematic content of the statement differs from the discourse during the Q&A session. The topics dominating the statement are persistent, structural, and strongly focused on standardised macroeconomic aggregates. Conversely, the Q&A often introduces new topics that were not addressed in the prepared text. Empirical research reveals that the first six most prevalent topics within a vast sample of press conference transcripts are predominantly driven by the Q&A segments. This suggests that a significant share of the added value of the press conference lies in the Q&A phase, which provides the public and market participants with detailed explanations regarding the ECB’s decisions in real time [24].

1.1.2 Other Communication Channels

While the MPS and the accompanying press conference represent the focal point of the ECB’s communication strategy, they are complemented by a broader set of publications, reports, and public engagements. To continuously influence the yield curve, anchor long-term expectations, and fulfill its mandate of transparency, the ECB applies several supplementary communication channels in the weeks between formal GC meetings. This *inter-meeting communication* is an effective mechanism for managing expectations between meetings and for addressing structural complexities and the mechanics of

financial transmission. These details cannot be fully addressed in a single monthly press conference [17].

The Economic Bulletin is a comprehensive analytical publication that explains the reasons behind the MP decisions of the ECB’s Governing Council. Published eight times a year—generally two weeks after the corresponding MP meeting—the Bulletin contextualises the Governing Council’s decisions against the detailed backdrop of prevailing economic, financial, and monetary conditions [7].

The Monetary Policy Accounts represent a significant transparency reform. The ECB began publishing regular accounts of the Governing Council’s MP meetings in early 2015, aligning its institutional practices with those of the US FED and the Bank of England. These accounts are typically released four weeks after the corresponding meeting [8].

The documents offer a detailed overview of the Governing Council’s internal deliberations. They outline the delicate balance of arguments, the range of risk assessments considered by various members, and the different perspectives on the economic outlook. By documenting the full range of analytical debates, the accounts help market participants understand the central bank’s internal response function and the specific macroeconomic thresholds that might trigger future policy interventions [5, p. 26–27].

As the ECB’s communication channels have multiplied to foster transparency, the volume of textual data generated by the central bank has expanded substantially. While human analysts can interpret individual statements, the sheer scale and frequency of this qualitative information necessitate computational approaches to systematically extract the monetary policy signals. This demand for automated, large-scale text analysis motivates the application of NLP.

1.2 Natural Language Processing in Finance

The growing availability of digital information has transformed the methodology of financial and macroeconomic research. Historically, empirical finance has relied almost exclusively on structured, quantitative data. However, language has always shaped financial markets, and unstructured textual data—ranging from earnings call transcripts to central bank MPS—contains critical information about the future. NLP serves as a technological linkage, enabling researchers and practitioners to systematically extract structured, machine-readable signals from these vast corpora of unstructured text [15]. In the context of MP, NLP algorithms are used to quantify the qualitative narratives of central bankers, capturing nuances in market psychology and institutional sentiment.

1.2.1 Concept of Sentiment Analysis

Sentiment analysis is the computational extraction of subjective information and emotional states from text, with the central task of classifying a passage’s underlying *polarity* [30]. In its canonical formulation the polarity is mapped onto a positive–neutral–negative axis, where *positive* denotes affirmative or optimistic framing and *negative* denotes adverse or pessimistic framing.

Early applications in finance relied on dictionary-based approaches, which measure sentiment by aggregating the frequency of predefined terms. A common metric in this tradition is a normalised difference between positive and negative word counts. However, as noted by Loughran and McDonald [23], standard baseline dictionaries routinely misclassify common financial terminology; the word *liability*, for example, is negative in general parlance but serves as a neutral, descriptive accounting term in financial filings. This mismatch motivated the development of domain-specific lexicons to accurately measure sentiment in corporate and macroeconomic contexts [23].

As deep learning matured, the methodology of sentiment quantification has shifted from counting isolated words to processing contextual probabilities. To facilitate econometric analysis, researchers transform these discrete class probabilities into a continuous *Sentiment Score* (S). While the technology has evolved, the logic of aggregation remains similar to classic sentiment indices [13]. As defined in Equation 1.1, the score is calculated as the difference between the probability assigned to the positive (or hawkish) class and its negative (or dovish) counterpart:

$$Score = P_{\text{positive}} - P_{\text{negative}}, \quad (1.1)$$

where $P \in [0, 1]$ represents the softmax probability assigned by the model to each respective label. This transformation maps the model’s probabilistic output onto a linear scale bounded by -1 and $+1$. A value of zero signifies either a balanced or a neutrally classified passage. The two classifiers used in this thesis (introduced below) emit a binary positive/negative softmax: passages that the model assigns to a third *neutral* class contribute zero by construction, attenuating the aggregate signal toward zero when a press conference consists predominantly of procedural or descriptive language.

The transition from static lexicons to transformer-based architectures, such as BERT (Bidirectional Encoder Representations from Transformers), addresses the inability of bag-of-words models to capture context-dependent relationships. Unlike methods that process words in isolation, BERT evaluates each token in relation to all others in a sentence, allowing for a nuanced treatment of syntax, negation, and complex economic narratives.

However, the linguistic requirements of central bank analysis nevertheless necessitate further specialisation. General-purpose models trained on Wikipedia or common

literature struggle with the technical and coded lexicon used by monetary authorities [12]. This limitation motivated the development of domain-adapted sentiment classifiers, of which two are employed in the present work. **FinBERT** [2] is a BERT-base model further pre-trained on Reuters’ TRC2 financial-news corpus and fine-tuned for sentiment on the Financial PhraseBank, and it represents the standard for general financial sentiment. **CentralBankRoBERTa** [25] is fine-tuned by Pfeifer and Marohl on a corpus of central bank speeches and reports and represents a further narrowing of the training distribution toward the institutional vocabulary of monetary authorities. Both models output a binary positive/negative polarity in the sense of Equation 1.1; they differ in the domain of their pre-training corpus, not in their classification objective. Their differences are summarised in Table 1.2.

Table 1.2: Evolution of NLP models in financial and central bank contexts.

Model	Training Corpus	Linguistic Nuance
BERT	General (Wikipedia, BookCorpus)	High (general context)
FinBERT	Reuters TRC2 + Financial PhraseBank	Domain-specific (financial)
CBRoBERTa	ECB/FED speeches and reports	Policy-specific (monetary)

A conceptually distinct strand of the literature seeks to classify not the *emotional* tone of central bank communications but their *policy stance*: whether a passage signals a tightening (“hawkish”), an easing (“dovish”) or no change (“neutral”) in the monetary stance. Stance classification differs from sentiment analysis along two dimensions. First, its label space is operational rather than affective: a sentence such as “*inflation is far too high*” carries a strongly negative emotional tone yet implies a hawkish policy stance, because elevated inflation is the standard precondition for tighter policy. Second, the construction of stance training sets requires manual annotation by domain experts using documented annotation guides, since the mapping from text to policy implication is not recoverable from sentiment cues alone. State-of-the-art stance classifiers for central bank discourse include the FOMC-trained RoBERTa-large model of Shah et al. [26] and the multi-bank extension of Shah, Sukhani, Pardawala et al. [27], the latter of which provides a model fine-tuned specifically on ECB communications.

These models are released under gated access on the Hugging Face Hub. As access could not be obtained within the timeframe of this study, the present analysis is restricted to sentiment classifiers, and the resulting signal should be interpreted as a measure of the *affective tone* of ECB discourse rather than its policy stance. Extending the pipeline to incorporate the stance dimension is identified in Chapter 4 as a priority for future work.

1.2.2 Zero-Shot Classification and the NLI Mechanism

While the polarity score provides a measure of the tone of the discourse, it does not identify the subject of that tone. A hawkish signal regarding inflation has vastly different market implications than a hawkish remark regarding structural reforms. Sentiment analysis is therefore often complemented by thematic categorisation.

Traditionally, the thematic analysis of central bank communication was dominated by probabilistic generative models, most notably Latent Dirichlet Allocation (LDA) [3]. While useful for broad explorations, these “bag-of-words” approaches suffer from significant limitations, specifically an inability to capture semantic context and the requirement for users to pre-specify a fixed number of topics without a deeper understanding of linguistic nuance. To overcome these limitations, the field of financial NLP has increasingly shifted toward neural architectures using transformers and Natural Language Inference (NLI) frameworks [31].

This architectural shift enables *zero-shot classification*: the technique allows a model to categorise text into user-defined labels without any task-specific supervised training. The procedure recasts classification as a logical relationship between a *premise* (the original central bank passage) and a *hypothesis* (a template representing the candidate label, such as “*This passage discusses monetary policy.*”) [31].

To derive a confidence score for a specific category, the model’s raw output logits for logical entailment and contradiction are processed through a softmax to yield the probability $P(c_i)$ for each candidate label c_i . In the implementation adopted in this work, the BART-large architecture pre-trained on the MultiNLI corpus [22] provides the entailment backbone, and the neutral logit is discarded prior to normalisation:

$$P(c_i) = \frac{\exp(l_{entail})}{\exp(l_{entail}) + \exp(l_{contra})}, \quad (1.2)$$

where l_{entail} and l_{contra} represent the model’s raw logits for entailment and contradiction, respectively. This mathematical framework allows the resulting thematic weight to reflect the relative strength of the logical link between the bank’s statement and the defined hypothesis.

By combining domain-specific sentiment extraction with zero-shot thematic classification, the qualitative narratives of central bankers are transformed into structured, time-series data. However, to evaluate whether these linguistic signals actually influence financial markets, they must be benchmarked against observable economic variables. This requires an econometric framework capable of quantifying both the official policy stance and the high-frequency market reactions.

1.3 Measuring Monetary Policy

To fill the gap between qualitative NLP outputs and econometric modelling, it is essential to define the quantitative metrics that represent the monetary policy stance and market expectations. This section details the primary indicators used as targets for predictive modelling, ranging from official policy rates to structural market shocks. Table 1.3 summarises the four indicators that recur throughout the empirical chapters.

Official Policy Rates and the Shadow Rate The primary instrument of the ECB’s monetary policy is the Main Refinancing Operations (MRO) rate, which sets the cost of liquidity for the banking system. While the MRO rate is the explicit signal of the policy stance, its effectiveness as a continuous variable is limited when interest rates reach the ZLB.

To measure the stance when nominal rates are constrained, researchers employ the *Shadow Rate*. As proposed by [19] and operationalised for the euro area by Wu and Xia [29], the shadow rate is a model-based construct that translates unconventional policy measures, such as large-scale asset purchases and forward guidance, into a single interest rate equivalent. Unlike the MRO, the shadow rate can move into negative territory, providing a more granular metric of monetary conditions during periods of intensive central bank intervention.

Market Expectations and OIS Volatility Real-time market reactions to central bank communication are primarily captured using OIS. An OIS is a financial derivative where a fixed interest rate is exchanged for a floating overnight rate (specifically the €STR in the Eurozone) over the duration of the contract. Since there is no exchange of principal and the floating leg is based on an overnight rate, OIS rates are considered a “risk-free” proxy for the market’s expectation of the future path of monetary policy.

In the context of the *EA-MPD* dataset [1], the focus is on the *change* in these rates (ΔOIS) during the press conference window. The dataset provides a term structure of expectations ranging from one month to ten years. Each segment serves a distinct analytical purpose: the short end (1M to 6M) reflects expectations regarding near-term policy rates, the mid-curve (1Y to 2Y) captures the effectiveness of forward guidance, and the long end (5Y to 10Y) acts as a gauge for long-term inflation anchoring.

Beyond the directional shifts in rates, their intraday volatility during the announcement serves as a critical proxy for market uncertainty. A divergence between the IS and the spontaneous Q&A session—often referred to as a “communication mismatch”—typically manifests as heightened volatility. This indicates that the market is struggling to price in conflicting signals, making uncertainty a quantifiable dimension of the central bank’s communication effectiveness.

Table 1.3: Key Monetary Policy Indicators used as empirical targets.

Indicator	Economic Significance
MRO Rate	Direct policy signal and cost of liquidity.
Shadow Rate	Policy stance proxy in ZLB environments [29].
OIS Rates (1M–10Y)	Forward-looking expectations across the yield curve.
<i>PC1</i> (Shocks)	Aggregated policy surprise [18].

Structural Shocks and Principal Component Analysis The “pure” impact of a policy announcement is often clouded by the information effect, where markets react to the central bank’s economic outlook rather than the policy change itself. To isolate the intended policy signal, [18] and [1] utilise structural decomposition. By observing the co-movement of interest rates and stock prices, researchers separate these shocks into “pure monetary” and “information” components. To simplify the multi-dimensional nature of interest rate moves across the yield curve, Principal Component Analysis (PCA) is applied to the high-frequency Δ OIS vector. The First Principal Component (*PC1*) is extracted to represent the overarching shift in expectations, as shown in Equation 1.3:

$$PC1_t = \sum_{j=1}^n \omega_j \cdot \text{Shock}_{j,t}, \quad (1.3)$$

where ω_j represents the PCA loading for the structural shock at maturity j . This *PC1* serves as a robust target variable for the machine-learning models of Chapter 3, as it captures the dominant policy surprise while filtering out idiosyncratic noise from individual maturities.

Chapter Synthesis The institutional structure of central bank communication, the computational extraction of sentiment via NLP, and the quantitative measurement of monetary policy shocks form the basis of this thesis. By mapping the qualitative shifts in the ECB’s discourse onto the structural changes in the OIS yield curve, the methodology of Chapter 2 details the empirical framework used to isolate the trading value of monetary policy signals, and the empirical findings of Chapter 3 evaluate it against the financial-market benchmarks defined above.

Chapter 2

Data and Methodology

This chapter outlines the data-collection framework and the implementation of NLP and machine-learning models. The data pipeline begins with the automated scraping and text processing of the ECB’s monetary-policy statement transcripts. The extracted and processed text is then stored in a custom SQLite database, alongside sentiment and topic metadata produced by BERT-family models and a zero-shot topic classifier. This architecture ensures data integrity and provides a robust foundation for the subsequent econometric analysis of central bank communication against financial-market variables.

2.1 Data Scraping and Preprocessing

Although datasets of ECB communications are publicly available, most sources provide uniform text blocks that do not distinguish the IS from the Q&A session, which limits their value for fine-grained sentiment analysis. A custom scraping pipeline is therefore required to isolate the responses of ECB officials from journalists’ questions and extract MP signals free from the sentiment bias of journalistic questioning. The first phase of the pipeline retrieves the transcript hyperlinks from GC press conferences and parses their content into two logical segments stored in a structured format.

After downloading the raw HTML, the pipeline separates the IS from the Q&A session using custom regular expressions that target structural markers such as “**My first question...**” or “**Transcript of the questions**”. Within the Q&A segment, the same mechanism distinguishes the responses of ECB officials from the journalists’ questions. The extracted text, together with dates and source URLs, is temporarily stored in Pipe-Separated Values (PSV) format.

In total, the pipeline retrieves **289 transcripts**, covering all press conferences between June 1998 and October 2024.¹ All transcripts share the same HTML layout,

¹https://www.ecb.europa.eu/press/press_conference/monetary-policy-statement/html/index.en.html

with content concentrated in <p> tags and headings of varying levels. The parsed output forms our **base dataset**, subsequently refined using the `nltk`, `re`, and `pandas` Python libraries.

The structural separation alone does not guarantee a clean signal: ceremonial greetings (e.g., “*Good afternoon...*”), procedural transitions, and short conversational fillers (e.g., “*Yes.*”) carry no MP information yet distort the sentiment baseline. To address this, the cleaning pipeline combines two filters: a Regex matcher that removes a list of pre-defined sentences, and a length-based heuristic that discards paragraphs shorter than 100 characters. The latter effectively eliminates section headers and short fillers without affecting substantive policy statements. Illustrative cases are presented in Table 2.1.

Table 2.1: Illustrative examples of noisy paragraphs excluded from the dataset.

Type	Example
Greeting	“Good afternoon, ladies and gentlemen.”
Procedural	“The Vice-President and I are very pleased to welcome you to our press conference.”
Closing	“We are now at your disposal for questions.”
Conversational	“Yes.”

2.1.1 Splitting the Q&A Section into Questions and Answers

A second structural challenge arises within the Q&A segment itself: unlike the continuous IS monologue, the dialogue between journalists and the ECB President requires explicit speaker attribution, which the HTML mark-up does not provide consistently. At least three distinct conventions for distinguishing questions from answers coexist in the corpus and evolve over time, as illustrated in Figure 2.1.

Since each question and each answer occupies its own paragraph, Q&A turns are stored row-wise in the PSV file introduced in Section 2.1. The resulting dataset has five primary features: **statement_id** (unique identifier), **date**, **url**, **intro** (the prepared introductory statement), and **qa** (the unscripted dialogue).

Within each row, the *intro* and *qa* fields are tab-delimited at paragraph level to preserve the internal structure of the conference. The *qa* field additionally wraps text segments in square brackets so as to retain the original bold and italic markers, which the ECB employs to flag a journalist’s question against the President’s answer (Figure 2.1). This convention is not universal: some transcripts instead prefix each turn with an explicit label such as **Question:** or the speaker’s name, followed by unformatted text.

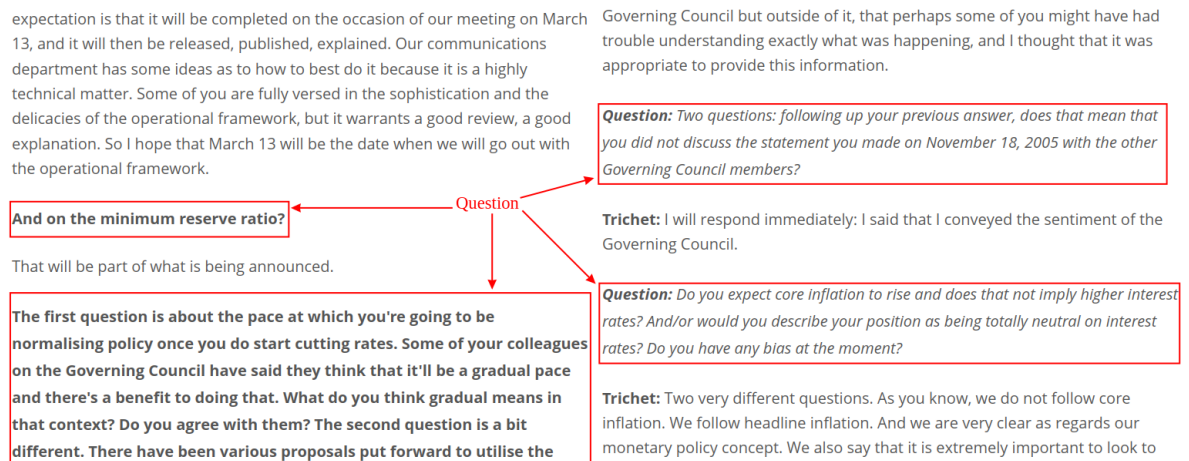


Figure 2.1: Evolution of formatting styles in ECB transcripts. The 2005 transcript (right) uses explicit speaker labels, whereas the 2024 version (left) relies on font styling (bold/italic) to distinguish between speakers.

2.2 Quantifying Central Bank Communications

Transforming qualitative narratives into quantitative market predictors requires a systematic refinement of the base dataset assembled in Section 2.1. The pipeline introduced in this section first segments each transcript into model-ready chunks and then, in parallel, applies zero-shot topic classification and sentiment scoring before persisting both the standardised chunks and their derived scores in an SQLite database. The overall architecture is illustrated in Figure 2.2; its output consists of two specialised collections corresponding to the IS and Q&A components, each preserving an audit trail back to every raw HTML paragraph.

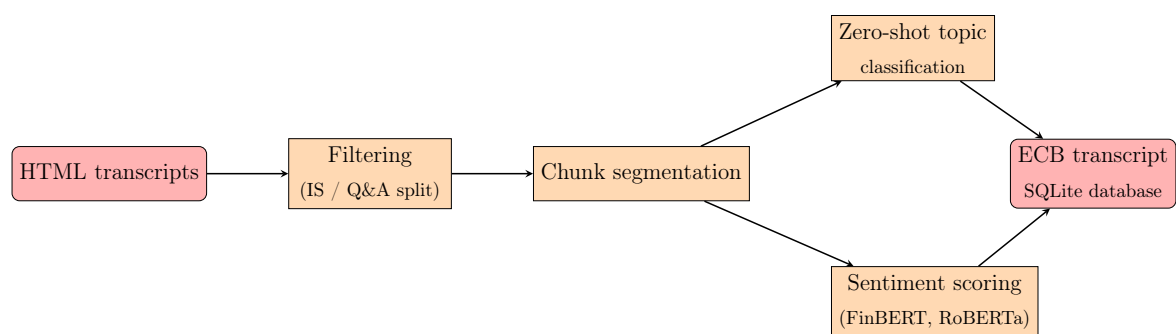


Figure 2.2: Data pipeline from raw HTML transcripts to the structured SQLite database. Topic and sentiment metadata are generated in parallel before being merged with the underlying chunks at the storage layer.

2.2.1 Text Segmentation

Sentiment classifiers operate on individual chunks rather than entire documents, and the BERT-based architectures employed in this work impose a hard 512-token input limit. To satisfy both constraints while preserving argumentative context, we set the chunk length to 200 words (≈ 260 – 340 subword tokens for typical central-bank prose). Paragraphs exceeding this limit are split at sentence boundaries using `nltk`’s sentence-boundary detector, whereas paragraphs shorter than 30 words are kept as single segments without further splitting. Each segment inherits its parent’s metadata (date, speaker, section type). The resulting word-count distribution, with both thresholds explicitly marked, is shown in Figure 2.3.

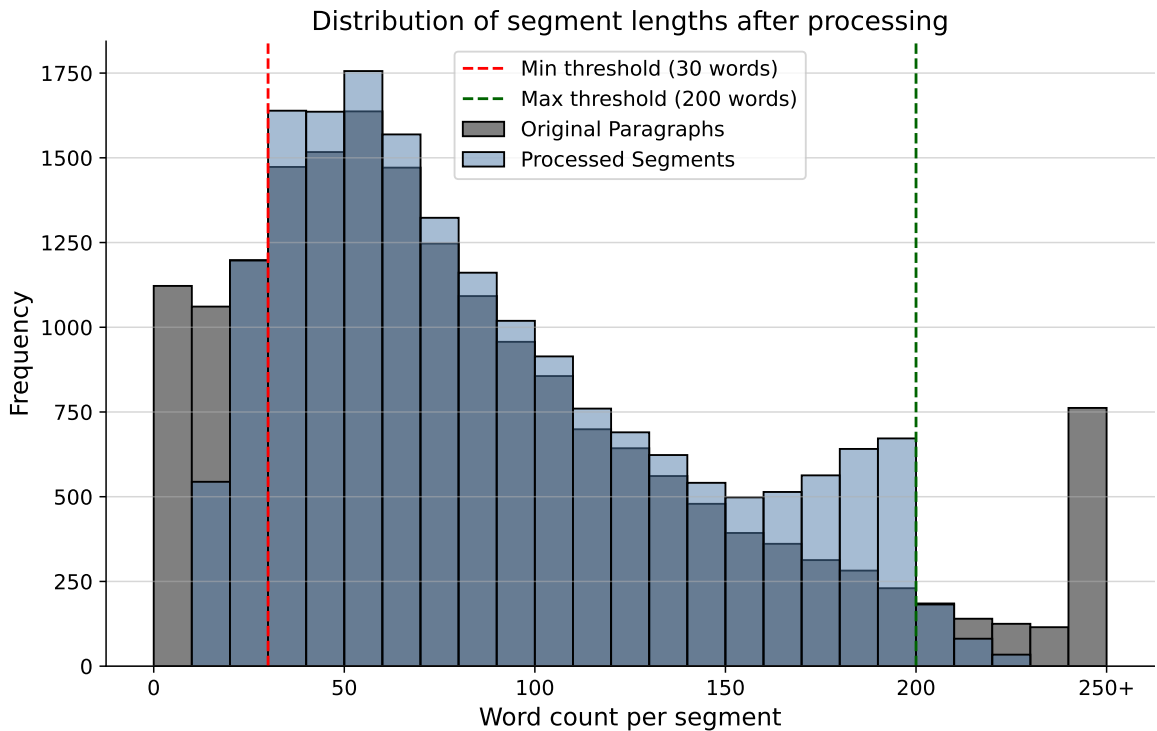


Figure 2.3: Impact of segmentation on text length distribution.

The 200-word threshold is the outcome of a sensitivity analysis across eight granularities (1, 50, 100, 150, 200, 250, 300, 350 words), reported in Table 2.2. The Pearson correlation between the FinBERT and CentralBankRoBERTa meeting-level means declines monotonically from $r = 0.727$ at single-sentence level to $r = 0.589$ at 350 words, while the within-meeting standard deviation $\bar{\sigma}$ stabilises beyond 200 words ($\Delta\bar{\sigma} < 0.001$ thereafter). Single-sentence chunks therefore maximise cross-model agreement but fragment multi-sentence economic arguments; 200 words is selected as the operating point that balances signal stability against context preservation. Granularities of 1 and 50 words are revisited as robustness checks in Chapter 3.

Table 2.2: Chunk-size sensitivity metrics. Cross-model r : Pearson correlation between FinBERT and CentralBankRoBERTa meeting-level means. $\bar{\sigma}$: mean within-meeting standard deviation of chunk-level scores. The bold row indicates the selected granularity.

Chunk size (words)	Cross-model r	$\bar{\sigma}$
1	0.727	0.506
50	0.716	0.509
100	0.676	0.474
150	0.645	0.451
200	0.616	0.444
250	0.602	0.441
300	0.599	0.441
350	0.589	0.441

2.2.2 Zero-Shot Topic Classification

The segmented chunks are now structurally clean but remain unlabelled. Traditional dictionary-based approaches cannot handle context such as negations or complex sentence patterns, and training a domain-specific topic-modelling transformer was infeasible given the absence of a pre-labelled ECB corpus and the time constraints of this project.

We therefore adopt **zero-shot classification**, which assigns arbitrary text to a pre-defined set of labels without label-specific training. The chosen backbone is *facebook/bart-large-mnli*, pre-trained on Natural Language Inference (NLI): each thematic label is treated as a hypothesis and the model returns its entailment probability against the input segment.

Iterating on the taxonomy. An initial ten-label taxonomy caused the classifier to spread its probability mass thinly and collapse heterogeneous segments into a single dominant category. Reducing the taxonomy to four broader themes and replacing short titles with descriptive hypothesis strings (Table 2.3) sharpened the entailment signal substantially. Because individual passages frequently touch on several themes at once (e.g., wage developments linked to the policy stance), the classifier is run in multi-label mode: every topic whose entailment probability exceeds $\tau = 0.45$ is assigned to the chunk, and the top-1 label is retained as a fallback when no label clears the threshold. The *Other* category isolates thematic noise—procedural exchanges and remarks without market-moving potential—so that its downstream effect on the sentiment signal can be tested explicitly in Chapter 3.

Table 2.3: Thematic labels and descriptive strings for Zero-Shot Classification.

Category	Model Input (Label String)
Monetary Policy	“inflation, price stability, interest rate decisions, monetary policy stance, financing conditions, bank lending, and market interest rates”
Economic Performance	“economic growth, GDP outlook, unemployment, labour market developments, macroeconomic risks, demand, consumption, and investment”
Fiscal & Structural	“government budgets, national debt, public spending, and structural reforms”
Other / Noise	“general greetings, purely political questions, unrelated remarks, climate change, or personal comments (excluding macroeconomic impacts)”

Robustness with a second NLI backbone. To rule out a model-specific artefact, the pipeline is replicated with a second backbone, *MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli-ling-wanli* [21], which differs from BART-MNLI both architecturally (encoder-only, disentangled attention) and in its substantially broader NLI fine-tuning corpus (MultiNLI, FEVER-NLI, ANLI, LingNLI, WANLI). Both backbones are applied to identical chunks under the same four-topic taxonomy and entailment threshold.

At the primary 200-word chunk size, top-1 agreement between the two backbones reaches only $\kappa = 0.155$ (raw agreement 27.8%, $n = 18,557$ chunks) [20]. DeBERTa-v3 systematically routes 60–72% of substantive ECB passages to the OTHER / IRRELEVANT class—a known consequence of its adversarial NLI fine-tuning raising the entailment bar beyond what our hypothesis strings can clear [21]. The comparison therefore does not constitute an independent robustness check: all downstream analyses use BART-large-mnli labels exclusively, and the DeBERTa results are reported as a transparency note.

2.2.3 Multi-Model Sentiment Extraction

With each chunk labelled, the next stage of the pipeline quantifies its tone. To guard against the idiosyncrasies of any single model, sentiment is extracted in parallel by two pre-trained transformers: **FinBERT** [2]—a BERT-base encoder pre-trained on Reuters

TRC2 and fine-tuned on the Financial PhraseBank— and **CentralBankRoBERTa** [25], a RoBERTa-base encoder fine-tuned on a curated corpus of central-bank speeches and reports. Both return a binary positive/negative polarity in the sense of Equation 1.1, but differ in encoder family, tokeniser, and fine-tuning data, providing two largely independent readings of the same text. Agreement on the sign of a chunk’s sentiment can therefore be treated as a model-invariant signal, whereas disagreement is itself informative and is analysed in Chapter 3. Representative outputs are shown in Table 2.4.

Table 2.4: Illustrative sentiment scores produced by the two classifiers on representative passages from the corpus. Scores are computed via Equation 1.1 from the softmax distribution returned by each model.

Passage	FinBERT	CB-RoBERTa
“Inflation is far too high and is expected to remain above our target for an extended period.”	−0.62	−0.71
“The economic outlook remains broadly balanced, supported by a resilient labour market and gradually easing financing conditions.”	+0.48	+0.05
“We are determined to bring inflation back down to two per cent over the medium term in a timely manner.”	+0.31	−0.18

2.3 Database Architecture

All artefacts produced by the pipeline—raw chunks, sentiment scores, and topic labels—are persisted in a custom *SQLite* database. The choice is deliberate: the dataset fits comfortably on a single machine, while the relational schema enforces the referential constraints that make every score and every label reproducibly traceable to a single original press conference.

The schema, illustrated in Figure 2.4, consists of seven normalised tables. The design separates raw text (`chunks`), model registries (`sentiment_models`, `topic_models`), the taxonomy (`topic_labels`), and the model-scored relations (`sentiments`, `topics`). Two design choices deserve emphasis. First, `topics` stores (`chunk`, `label`, `model`) triples together with the raw entailment probability `prob`, so that the multi-label threshold τ can be re-tuned post hoc without re-running the classifier and the BART/DeBERTa comparison of Section 2.2.2 can be performed against a common chunk store. Second, the `chunk_limit` column on `chunks` permits the simultaneous storage of multiple

segmentation granularities, so that the chunk-size sensitivity analysis of Section 2.2.1 runs against a single, internally consistent backing store.

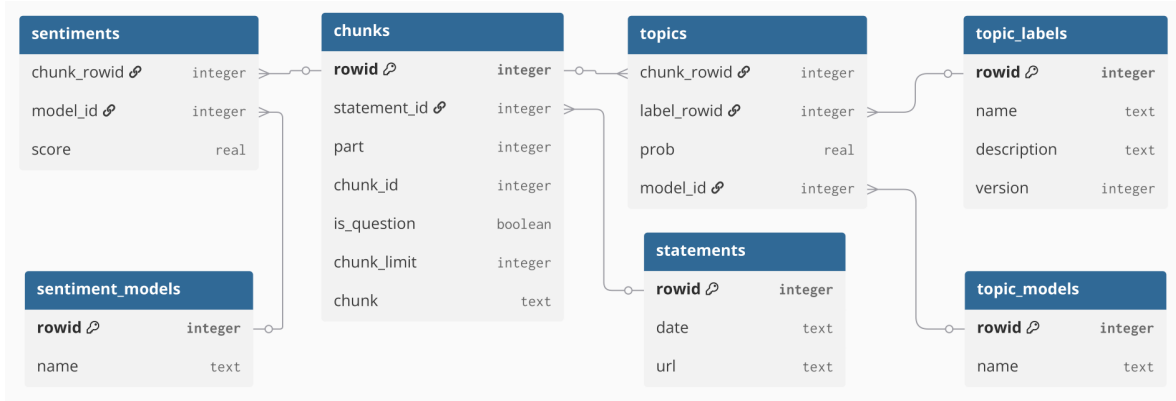


Figure 2.4: Physical ER diagram of the ECB transcript database (generated with *dbdiagram.io*), showing the seven tables, their columns, and their primary/foreign-key relationships.

All downstream analyses access the database through a thin Python query layer, which aggregates chunk-level scores into the meeting-level feature matrix described in Section 2.5.

2.4 Financial Market Data

To evaluate the informational value of the textual signal, the latter is confronted with a panel of financial indicators organised in three layers: high-frequency market surprises, official policy rates, and a derived structural shock.

High-frequency market surprises. The primary source of market expectations is the Euro Area Monetary Policy Event-Study Database (EA-MPD) of [1], from which we extract intraday changes in OIS rates at the 1-month, 3-month, 6-month, and 1-year maturities. The narrow event window around each press conference filters out unrelated macroeconomic news and thereby isolates the surprise component of ECB communication. The corresponding intraday change in the STOXX50 equity index is added to capture the broader risk reaction.

Official policy rates. The MRO rate and the Deposit Facility (DF) rate, sourced from the ECB money-market archives, represent the explicit policy signal. For periods in which nominal rates are constrained by the ZLB, the Wu–Xia shadow rate [29] is used as a proxy for unconstrained monetary conditions.

First principal component of the OIS surprises. To collapse the multi-dimensional OIS surprise vector onto a single target variable, we extract its first principal component, denoted **PC1**. PC1 captures the dominant level shift across the maturity spectrum and serves as a parsimonious summary of the policy surprise for the predictive analysis of Chapter 3.

The three layers are joined to the meeting-level feature matrix of Section 2.5 on the press-conference date, yielding the unified (date $\times$ feature) panel on which the empirical analysis operates.

2.5 Data Integration and Modular Feature Framework

The final stage of the methodology integrates the textual and financial data into a single feature matrix. Rather than committing to a fixed table layout, the pipeline is implemented as a **modular framework** whose granularity is selected at run time, so that the same codebase supports hypotheses ranging from aggregate market reactions to the granular dialogue between journalists and the central bank.

2.5.1 Configurable Granularities

The framework exposes three nested levels of granularity. The **Global** configuration treats each transcript as a single unit, collapsing all chunks into one sentiment summary per meeting. The **Sectional** configuration splits the discourse into the prepared IS and the spontaneous Q&A session, isolating the prepared narrative from the live exchange. The **Thematic** configuration additionally cross-cuts the data with the four macroeconomic topic labels of Section 2.2.2, producing a multi-dimensional view of the President’s stance on each policy area. A finer fourth mode separates **journalist questions from official answers**, providing the substrate for the divergence analysis of Section 3.2.3.

2.5.2 The Chunk-to-Meeting Aggregation Operator

The relational database of Section 2.3 stores sentiment and topic outputs at the granularity of the individual chunk, whereas econometric analysis requires a feature matrix indexed by press-conference date. Let $\mathcal{C}_{t,m,s,\tau}$ denote the set of chunks observed at press conference t , scored by sentiment model $m \in \{\text{FinBERT}, \text{CB-RoBERTa}\}$, belonging to section $s \in \{\text{IS}, \text{QA}\}$, and classified under topic $\tau \in \{\text{MP}, \text{EP}, \text{FS}, \text{OI}\}$. For each non-empty cell, the chunk-level scores are summarised by the four-dimensional moment vector

$$\mathbf{f}_{t,m,s,\tau} = (\bar{S}_{t,m,s,\tau}, \sigma_{t,m,s,\tau}, S_{t,m,s,\tau}^{\min}, S_{t,m,s,\tau}^{\max}). \quad (2.1)$$

The mean captures the central tendency of the tone, the standard deviation captures within-meeting dispersion, and the extrema preserve the strongest hawkish and dovish signals without requiring the analyst to specify a quantile threshold *ex ante*. The vector is computed across every non-empty cell of the four-way product $\{t\} \times \{m\} \times \{s\} \times \{\tau\}$, yielding a sparse but information-rich representation of each press conference.

Treatment of the Q&A section. Within Q&A cells, aggregation is restricted to the President’s answers: chunks flagged as journalist questions (`is_question = 1`) are excluded before Equation 2.1 is applied, so that the tone of the press corps does not contaminate the policy signal. The excluded journalist chunks are retained in the database and revisited in Section 3.2.3, where the gap between journalist framing and the ECB’s answers is treated as an object of analysis in its own right.

2.5.3 Missing Cells and Final Merge

The Cartesian product $\{t\} \times \{m\} \times \{s\} \times \{\tau\}$ contains cells for which $\mathcal{C}_{t,m,s,\tau}$ is empty—most commonly when a routine conference, dominated by inflation discussion, contains no chunks classified as `FISCAL_AND_STRUCTURAL`. To avoid the cross-sectional leakage that column-mean imputation would introduce, each missing entry is instead filled by carrying the most recent non-missing value of the same series forward in time. This encodes the economically natural prior that the silence of the central bank on a topic does not reset market beliefs about its most recently communicated tone on that topic. Observations preceding the first non-missing value of any series are excluded from the analysis sample, as no carry-forward is feasible without prior information.

The meeting-level feature matrix is then inner-joined on the press-conference date with the financial-market panel introduced in Section 2.4; meetings lacking either the textual record or the high-frequency market window are dropped at this stage. The surviving sample constitutes the unified (date $\times$ feature) matrix on which the empirical analysis of Chapter 3 operates. That chapter subjects these signals to four progressively more demanding tests: their internal coherence (Sections 3.1–3.1), their sectional and thematic structure (Section 3.2), their co-movement with the monetary-policy cycle (Section 3.3), and their predictive value for the direction of ECB rate decisions (Section 3.4).

Chapter 3

Results

This chapter reports the empirical findings of the thesis. The analysis proceeds in four arcs of increasing demand on the data. Section 3.1 establishes that the two sentiment classifiers produce internally coherent signals and shows that those signals track major historical episodes of monetary-policy stress. Section 3.2 dissects the press conference into its prepared and spontaneous components, demonstrating that the IS and the Q&A session carry structurally different information. Section 3.3 relates the aggregate sentiment series to the ECB’s policy-rate cycle, including an event-study of rate decisions. Section 3.4 closes the loop by asking whether the textual signal carries predictive value for the direction of the next rate decision, comparing four classifiers of increasing complexity (a logit baseline, a sentiment-only logit, an augmented logit, and a random forest) and identifying the regimes in which sentiment-derived features add information beyond the rate history alone.

3.1 Aggregate Sentiment: Classifier Properties and Historical Coherence

Before examining the granular structure of ECB communication, we must confirm that the two sentiment classifiers produce internally coherent signals. Figure 3.1 plots the six-month centred moving average of both models together with ± 1 standard-deviation bands.

A persistent vertical gap separates the two series: FinBERT is mildly positive ($\bar{x}_{\text{IS}} = +0.208$) while CentralBankRoBERTa is anchored below zero ($\bar{x}_{\text{IS}} = -0.140$). The gap reflects neither a pipeline error nor a directional disagreement—it is a calibration offset inherited from the respective fine-tuning corpora (Section 1.2.1; Table 1.2). Table 3.1 quantifies the distributional properties of both classifiers.

Three patterns stand out. First, the IS is substantially more volatile than the QA for both models: the IQR is three times wider in FinBERT (0.290 vs. 0.094) and 50% wider

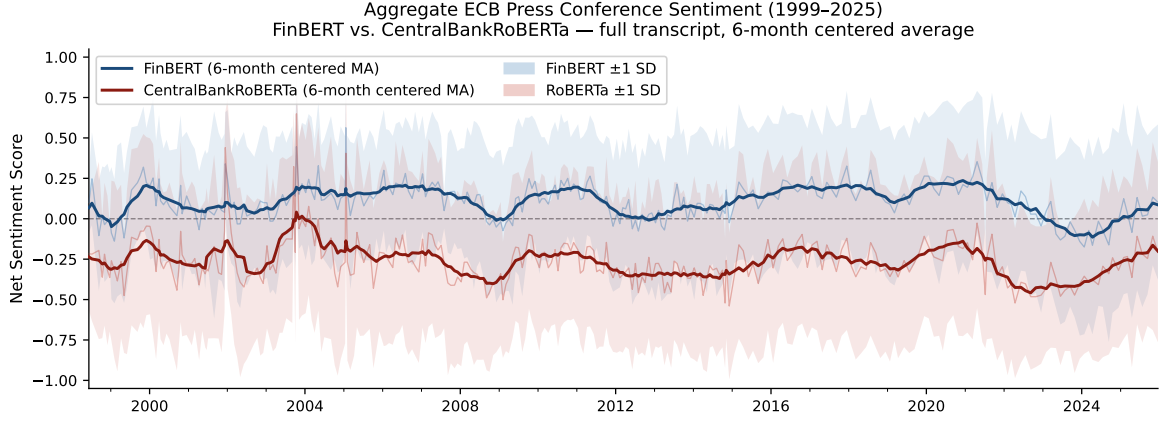


Figure 3.1: Aggregate ECB sentiment, six-month centred moving average (unitless; $n = 289$ IS, 281 Q&A meetings). Shaded: ± 1 SD

Table 3.1: Descriptive statistics of the net sentiment score, 1998–2025. IS: Introductory Statement ($n = 289$); QA: Q&A session ($n = 281$); IQR: interquartile range.

Model	Section	Mean	Std	Min	Max	IQR
FinBERT	IS	+0.21	0.22	−0.35	+0.71	0.29
FinBERT	QA	+0.05	0.07	−0.15	+0.27	0.09
CB-RoBERTa	IS	−0.14	0.19	−0.62	+0.65	0.26
CB-RoBERTa	QA	−0.34	0.12	−0.56	+0.01	0.17

in CentralBankRoBERTa (0.258 vs. 0.170). Second, the CentralBankRoBERTa QA distribution barely breaches zero at its maximum (+0.010), indicating that spontaneous communication never produces unambiguously positive sentiment on the central-bank scale. Third, the calibration offset is not uniform across sections: 0.35 in the IS against 0.39 in the QA, showing that the small gap between classifiers depends on communicative register.

The IS–QA Pearson correlations within each model (Table 3.2) are positive, moderate, and significant at the 0.1% level, confirming that both sections encode the same underlying policy stance at different intensities.

Table 3.2: Pearson correlation between IS and Q&A meeting-level mean sentiment ($n = 281$ matched meetings).

Model	Pearson r	p -value
FinBERT	0.243	< 0.001
CB-RoBERTa	0.256	< 0.001

A coherent pair of series can still be jointly wrong. To verify alignment with the

policy record, Figure 3.2 overlays the aggregate trajectory with seven historical anchor events.

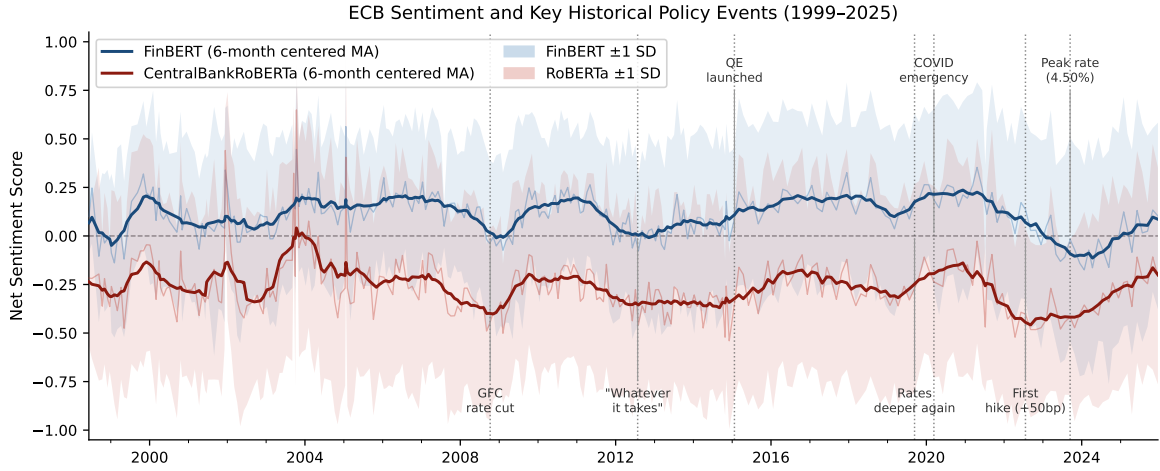


Figure 3.2: Aggregate ECB sentiment overlaid with seven anchor events. Dotted vertical lines mark the event date.

Both series fall sharply at the October 2008 Great Financial Crisis (GFC) rate cut, partially recover at Draghi’s “Whatever it takes” speech of July 2012, and drift further downward through the Quantitative Easing (QE) launch (January 2015) and deepening into negative territory (September 2019). The March 2020 pandemic emergency produces a second trough, followed by an unambiguous upswing ahead of the July 2022 liftoff; sentiment peaks around the September 2023 meeting at which the rate reached 4.50% and softens thereafter. The alignment across all seven events certifies that the signal is not artefactual; the more demanding co-movement tests with market variables are deferred to Section 3.3.

3.2 The Sectional Anatomy of Policy Communication

The headline series collapses the rich internal structure of a press conference into a single number. We now exploit the sectional split documented in Section 2.1.1 to ask whether the prepared and spontaneous components communicate different information and whether any structural gap between them carries market-moving content.¹

¹The raw corpus contains 289 press conferences. The cross-section analyses of Section 3.2 use $n = 272$ after the inner join with the EA-MPD market panel; the predictive analyses of Section 3.4 use $n = 277$ – 279 after lag-feature construction.

3.2.1 IS versus Q&A: Two Different Documents

Figure 3.3 plots the two sections separately across the full sample. Three regularities are immediately visible. First, IS sentiment dominates QA sentiment at almost every date for both models. Second, the IS series is markedly more volatile—the IQR ratio documented in Table 3.1 is approximately three to one. Third, the two series move broadly in the same direction at low frequency but diverge at higher frequencies, a pattern already quantified in Table 3.2: both classifiers explain less than 7% of the other section’s variance (FinBERT: $R^2 = 0.059$; CB-RoBERTa: $R^2 = 0.066$). Approximately 93 – 94 % of the Q&A’s variance is orthogonal to the IS.

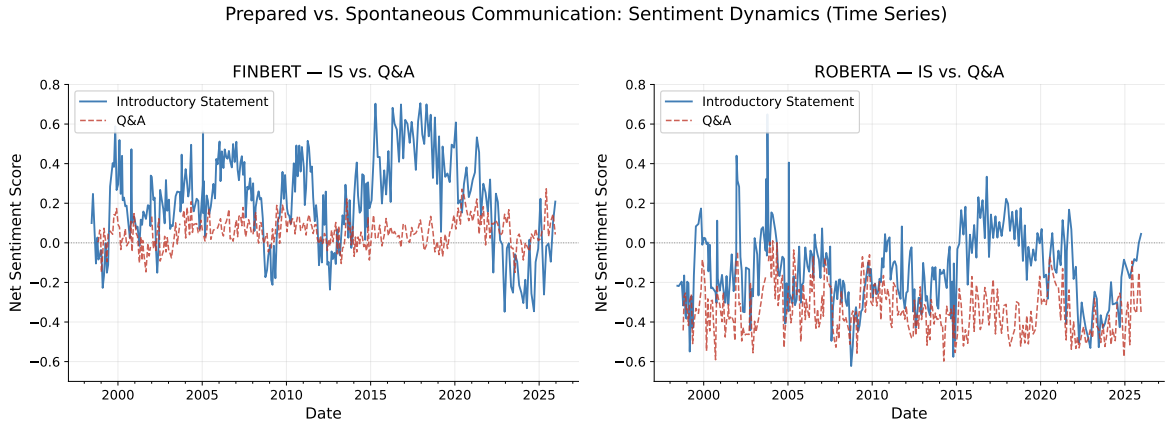


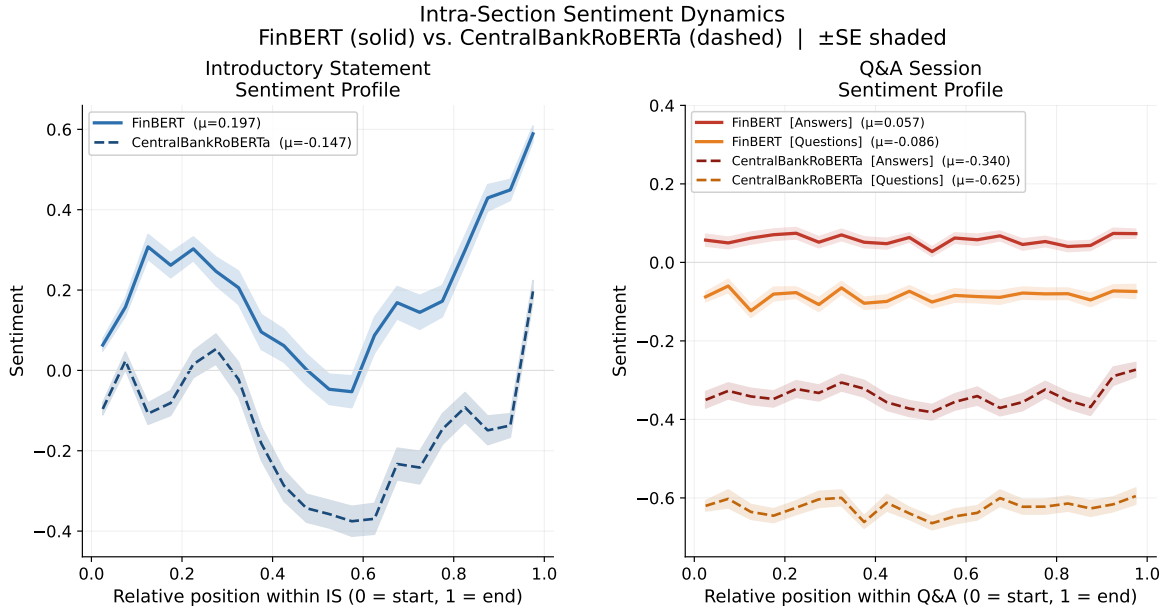
Figure 3.3: Introductory Statement versus Q&A sentiment over time. Left: FinBERT; right: CentralBankRoBERTa.

3.2.2 Internal Structure of the Press Conference

Because the data are segmented into short chunks, we can track how sentiment evolves *within* each section. Figure 3.4 bins the chunks by their relative position and averages across all meetings ($n_{IS} = 289$; $n_{QA} = 281$).

The IS profile traces a coherent three-act structure: it opens at a moderate level, dips between the one-fifth and two-fifths marks where the ECB catalogues downside risks, then rises sharply through the final passages where the policy decision and forward guidance are stated—a swing of approximately 0.4 sentiment units in FinBERT from trough to close. The Q&A profile, by contrast, is essentially flat: a sharp question lands as easily in the first exchange as in the last, and treating the Q&A as a single narrative document averages over a random sequence of topics rather than a coherent arc. This is the empirical justification for the chunk-level aggregation operator of Section 2.5.2.

The right panel also reveals a stable two-tier structure within the Q&A. ECB answers score consistently above journalist questions for both models, with mean gaps of 0.14 (FinBERT) and 0.29 (CentralBankRoBERTa). Figure 3.5 reports the CB-RoBERTa

Figure 3.4: Intra-section sentiment dynamics. Shaded: ± 1 SE.

gap, on which the asymmetry is more pronounced; the FinBERT counterpart shows the same sign but a smaller magnitude.

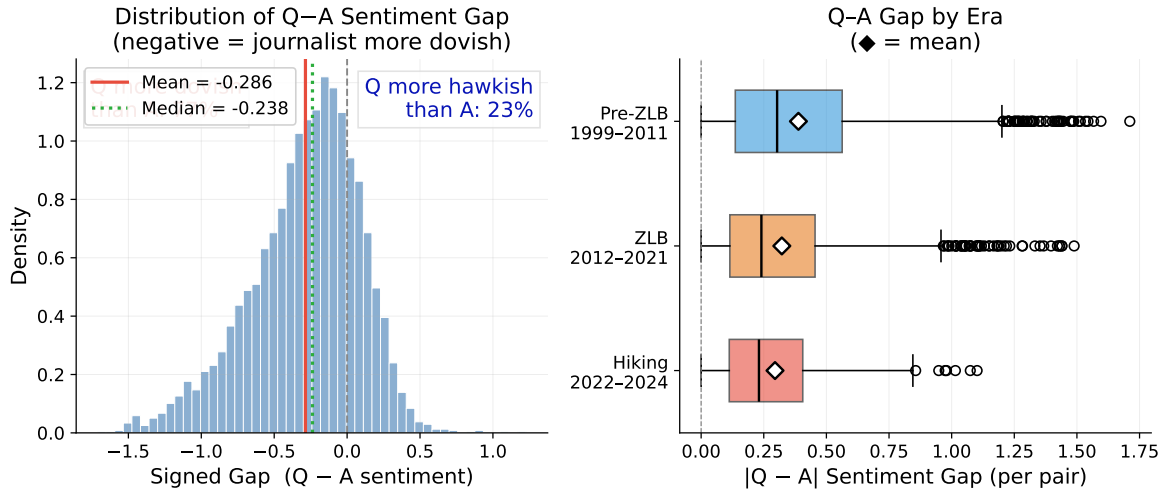


Figure 3.5: Journalist-minus-ECB sentiment gap (CentralBankRoBERTa). Left: full-sample histogram. Right: absolute gap by monetary regime.

The signed gap $S_Q - S_A$ has a mean of -0.286 and a median of -0.238 ; journalists are more dovish than the ECB in 77% of question-answer pairs. The right panel shows that the absolute gap is stable across eras, with mean values near 0.30 in both the Pre-ZLB and ZLB periods and 0.27 in the hiking cycle. This regularity reflects divergent communicative incentives: journalists are professionally rewarded for surfacing downside risks, while ECB officials are constitutionally bound to project measured stability. The gap is a structural feature of the discourse, not a measurement error; had the pipeline

not separated the two voices, the press corps' affective tone would have contaminated the policy signal by approximately 0.3 on the unit scale.

3.2.3 Does the Sectional Divergence Move Markets?

The two preceding subsections established that the IS and Q&A carry structurally different information and that the journalist–ECB gap is a stable institutional feature. A natural follow-up is whether either source of divergence carries market-moving content. Because three market variables are examined for each of two divergence measures, six Pearson correlations are tested simultaneously; we apply Holm–Bonferroni correction to the full family throughout and report both raw and adjusted p -values.²

Figure 3.6 plots the absolute IS–Q&A divergence against three market outcomes: the cross-maturity mean of $|\Delta\text{OIS}|$, the STOXX50 return, and the PC1 monetary surprise.

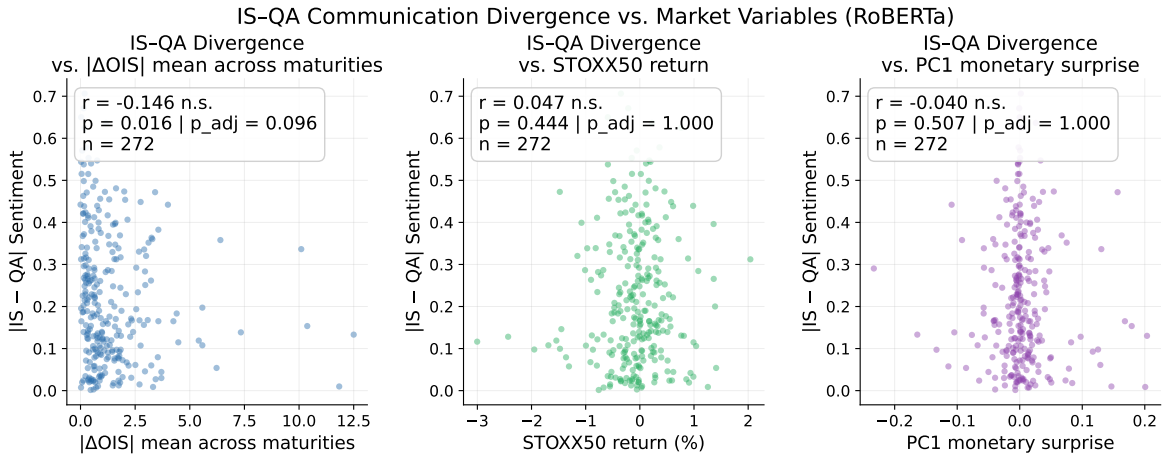


Figure 3.6: IS–Q&A sentiment divergence versus market variables ($n = 272$ press conferences). Each panel reports Pearson r , raw p , and Holm-adjusted p_{adj} ; significance stars reflect p_{adj} . Regression lines are shown for visual reference only.

The result is uniformly null after multiplicity correction. Neither the STOXX50 return ($r = 0.047$, $p_{\text{adj}} = 1.000$) nor the PC1 monetary surprise ($r = -0.040$, $p_{\text{adj}} = 1.000$) correlates meaningfully with the IS–Q&A divergence. A nominally significant negative association emerges with $|\Delta\text{OIS}|$ ($r = -0.146$, $p_{\text{raw}} = 0.016$), but this result does not survive Holm–Bonferroni correction ($p_{\text{adj}} = 0.096$, Bonferroni threshold ≈ 0.0083) and cannot be treated as a confirmed finding.

²Under the global null, six independent tests at $\alpha = 0.05$ yield an expected false-positive count of $6 \times 0.05 = 0.30$ and a probability of at least one spurious significant result of $1 - (0.95)^6 \approx 26\%$. The Holm–Bonferroni procedure controls the family-wise error rate while being uniformly more powerful than the plain Bonferroni correction [14]. Adjusted p -values are reproduced in the scatter-plot annotations of Figures 3.6–3.7 for transparency.

The *direction* of this association is nonetheless theoretically interpretable: press conferences in which IS and Q&A sentiment diverge most tend to produce the *smallest* OIS reactions. We label this directional pattern *communicative cancellation* — the hypothesis that conflicting signals within the same press conference prevent the market from cleanly updating its rate expectations in either direction. We treat this as a motivated conjecture rather than an established result and revisit it in the regime-dependent rolling analysis of Section 3.4.3, where the same channel is examined conditional on the prevailing monetary policy uncertainty regime.

Figure 3.7 tests whether the journalist–ECB sentiment gap within the Q&A carries analogous market content.

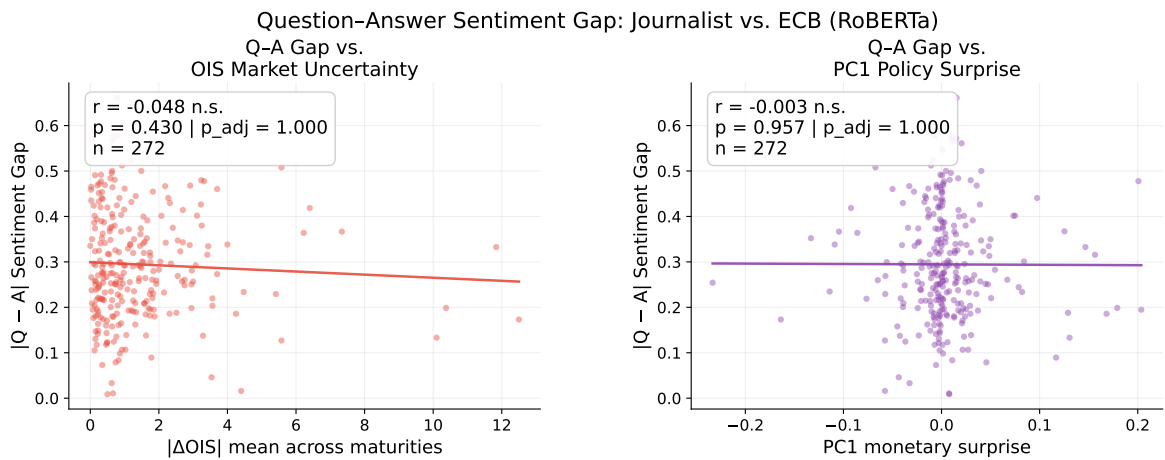


Figure 3.7: Internal Q–A sentiment gap versus market surprises ($n = 272$ press conferences). Annotation convention follows Figure 3.6. All three correlations are statistically indistinguishable from zero before and after family-wise correction.

All three Q–A gap correlations are firmly null at both levels ($p_{adj} \geq 0.764$ in every case; see Table 3.3). The near-flat regression lines in Figure 3.7 confirm the visual message. Markets appear to correctly anticipate the structural pessimism of the press corps and the measured caution of the ECB; these expected components are priced immediately and leave no residual correlation with realised market moves. It is the *deviation* from the institutionally expected baseline—not the level of the gap itself—that may carry market-moving content, a distinction testable only with event-study or high-frequency identification designs beyond the scope of this analysis.

Table 3.3 consolidates all six correlations and their family-corrected counterparts.

In summary, neither the IS–Q&A divergence nor the journalist–ECB gap moves markets in a statistically confirmed manner once multiplicity is accounted for. The directional evidence for communicative cancellation on OIS rate expectations is suggestive and carries a clear economic interpretation, but it requires corroboration in a regime-conditional framework before it can support strong inference.

Table 3.3: Pearson correlations between divergence measures and market variables with Holm–Bonferroni correction (family $\alpha = 0.05$, $n = 272$)

Outcome	Predictor	r	p_{raw}	p_{adj}	Decision
IS–QA divergence	$ \Delta\text{OIS} $	−0.146	0.016	0.096	Retain H_0
IS–QA divergence	STOXX50	0.047	0.444	1.000	Retain H_0
IS–QA divergence	PC1	−0.040	0.507	1.000	Retain H_0
Q–A gap	$ \Delta\text{OIS} $	−0.048	0.430	1.000	Retain H_0
Q–A gap	STOXX50	0.087	0.153	0.764	Retain H_0
Q–A gap	PC1	−0.003	0.957	1.000	Retain H_0

Note: Holm–Bonferroni correction applied simultaneously to all six tests. Plain Bonferroni threshold: $0.05/6 \approx 0.0083$. The nominally significant IS–QA vs. $|\Delta\text{OIS}|$ result ($p_{\text{raw}} = 0.016$) does not survive family-wise correction ($p_{\text{adj}} = 0.096$). All six null hypotheses are retained at the corrected level.

3.3 Sentiment and the Monetary Policy Cycle

The preceding sections established internal coherence and structural differentiation. The question now is whether the sentiment indices track the actual price of money: hawkish language should accompany rising rates and dovish language should precede cuts.

3.3.1 Co-Movement with Policy Rates

Figure 3.8 plots the smoothed sentiment series against the MRO rate and, in the ZLB window, against the Wu–Xia shadow rate [29].

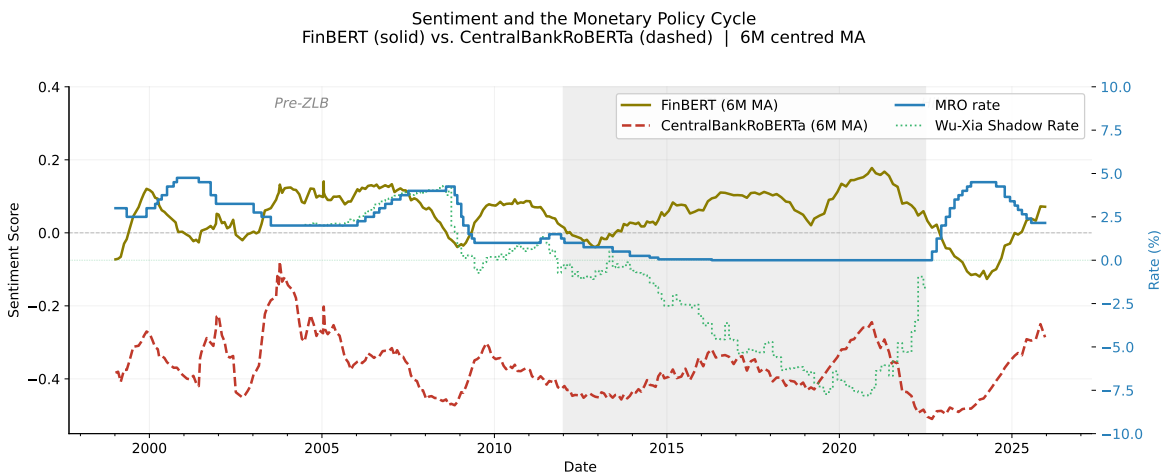


Figure 3.8: Six-month centred moving average sentiment versus the MRO rate and Wu–Xia shadow rate [29]. Shaded: ZLB era.

In the pre-ZLB window (1999–2011), sentiment and the MRO rate move broadly

in tandem, with sentiment leading rate adjustments by a small number of meetings. In the ZLB window (2012–2022)³, the MRO flatlines near zero while the shadow rate declines to approximately -7.5% by 2020. The sentiment series does *not* track this descent, remaining broadly flat—a consequence of the structural mismatch between hawk–dove vocabulary and the language of quantitative easing, discussed in Chapter 4. In the post-ZLB hiking window (2022–2025), sentiment turns visibly upward for both models and co-movement with the rate is restored.

3.3.2 Sentiment by Rate-Decision Type

Figure 3.9 plots the distribution of meeting-level IS sentiment separately for hiking, hold, and cutting meetings. Cutting meetings are the most negative for both models:

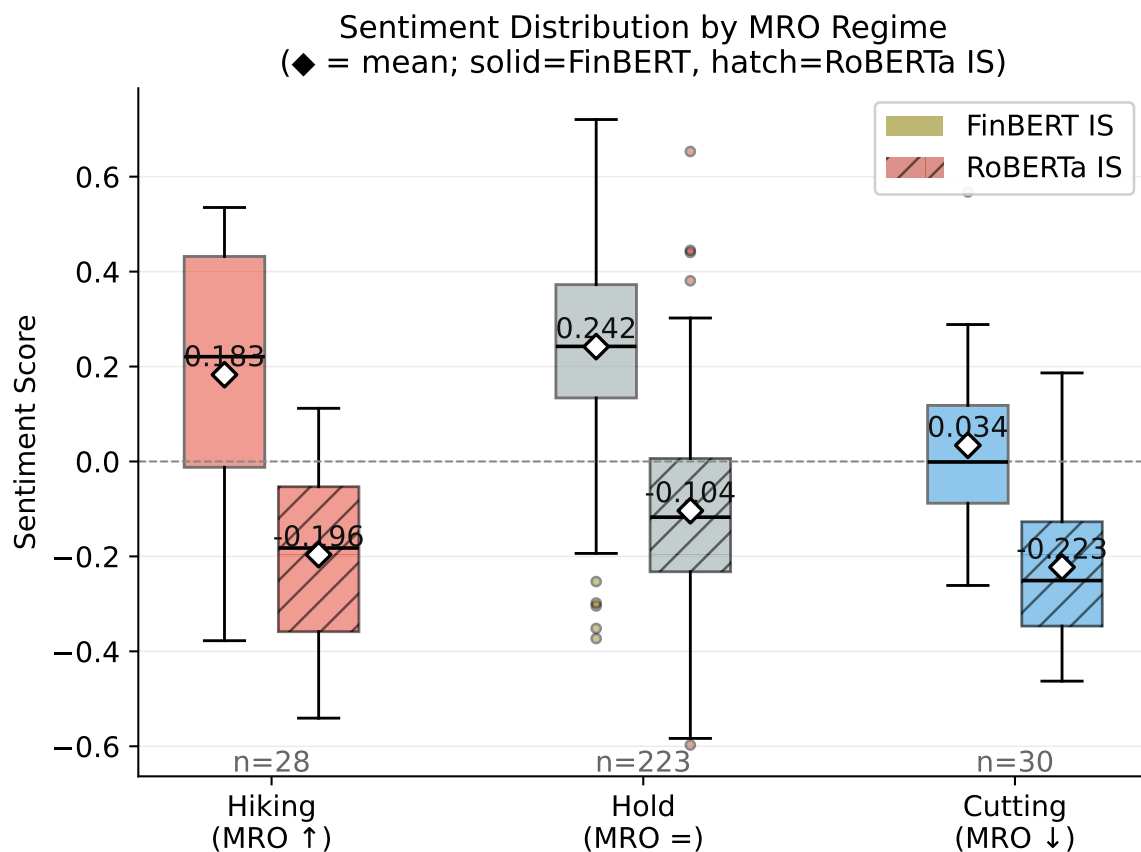


Figure 3.9: IS sentiment by rate-decision type

FinBERT $\bar{x} = 0.034$ versus $+0.242$ at hold meetings, and CentralBankRoBERTa $\bar{x} = -0.223$ versus -0.104 —the easing signal is detectable. The tightening signal

³The ZLB era is dated from January 2012, when the ECB’s shift to unconventional tools began following the sovereign-debt crisis and the Wu–Xia shadow rate [29] started to diverge from the MRO. The MRO itself reached zero in March 2016; the era ends in June 2022, the last meeting before the first conventional rate hike of 21 July 2022.

is weaker: hiking meetings are *less* positive than hold meetings for both classifiers (+0.183 vs. +0.242; -0.196 vs. -0.104), yielding Hold > Hiking > Cutting rather than the theoretically expected Hiking > Hold > Cutting. This is consistent with the ZLB-era contamination: the large hold cohort ($n = 223$) is dominated by pre-2022 meetings of measured stability, while the hiking cohort ($n = 28$) is drawn entirely from the 2022–2023 cycle in which alarm about inflation coexisted with hawkish policy intentions.

3.3.3 The Anatomy of Tightening versus Easing: An Event Study

Does the ECB telegraph rate changes in advance, and does it communicate differently when tightening versus easing? We align all rate decisions at lag = 0 and extract the sentiment trajectory in a ± 6 -meeting window. The analysis covers the pre-ZLB period (1999–2014; $n_{\text{hike}} = 15$, $n_{\text{cut}} = 15$) and the post-ZLB period (2022–2025; $n_{\text{hike}} = 10$, $n_{\text{cut}} = 8$); the ZLB era is excluded because rate changes are too rare to support meaningful window averages.

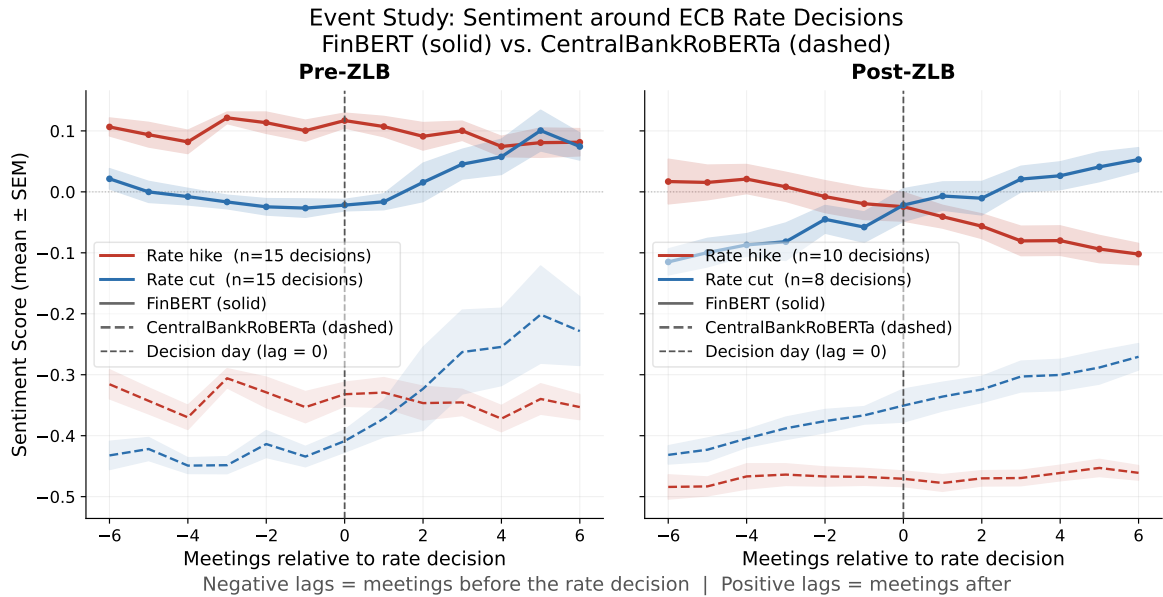


Figure 3.10: Sentiment in a ± 6 -meeting window around rate decisions.

The pre-ZLB panel shows a clear asymmetry. The hawkish profile (red) is fully established six meetings before the actual hike, forming a flat plateau near +0.10 in FinBERT. The dovish profile (blue) is qualitatively different: sentiment stays near zero through lag = -4 , dips modestly to ≈ -0.02 at the preceding meeting, then recovers to +0.1 within five meetings. Rate cuts in this period were predominantly emergency reactions that did not permit gradual preparation; the post-cut recovery reflects the ECB framing the action as stabilising.

The post-ZLB panel reveals a structurally different regime. The two FinBERT trajectories form a crossing X-shape at the decision meeting: the hiking profile starts near zero ($\approx +0.01$ at lag = -6) and declines to -0.10 by lag = $+6$; the cutting profile starts notably negative (≈ -0.12) and rises to $+0.05$. Both directions are telegraphed ahead of the policy action. The post-decision reversal—hikes followed by caution about over-tightening, cuts followed by confidence that inflation is under control—is consistent with the asymmetric distributional finding of Section 3.3.2. Sentiment leads the policy action, and the communication strategy differs fundamentally between eras, motivating the predictive analysis of Section 3.4.

3.4 Predictive Value of Sentiment

The preceding sections established that the sentiment series is coherent, structurally differentiated, and leads the policy rate around decision events. We now ask whether the textual signal predicts the *direction* of the next ECB rate decision once the rate history is controlled for. Section 3.4.1 characterises the lag-and-era correlation structure that motivates feature selection. Section 3.4.2 fits and compares four classifiers on the meeting-level rate-decision direction. Section 3.4.3 closes the chapter with an analysis of whether communication surprises move markets contemporaneously.

3.4.1 Cross-Era Predictive Correlations

Figure 3.11 traces the Pearson correlation between three selected sentiment features and the contemporaneous MRO change across horizons of zero to fifteen meetings (changed meetings only, $n \approx 54$). Two patterns emerge. First, the correlation is already small but significant at lag 0 ($r = 0.31$, $p < 0.05$ for the FinBERT overall mean); it rises to a pronounced peak at lag 4 ($r = 0.551$). Second, the profile is non-monotonic, with a secondary plateau through lags 6–10, indicating that sentiment carries predictive content at multiple horizons. The three features peak at different lags—FinBERT IS at lag 2 ($r = 0.352$), and CentralBankRoBERTa IS at lag 9 ($r = 0.409$)—so the textual signal is not a single-horizon lead.

Table 3.4 disaggregates the signal across monetary regimes. Three regularities emerge. First, correlations are positive in both the Pre-ZLB and Hiking regimes for almost every feature. Second, the ZLB column is sign-heterogeneous: extremum features are negative or near zero while level-mean features at lag 4 are weakly positive (range $+0.22$ – $+0.28$)—neither class tracks the shadow-rate accommodation reliably, for the reasons discussed in Section 3.3.1. Third, the four highest full-sample correlations belong to *within-meeting maxima* at a two-meeting lag; however, this signal is concentrated in the Pre-ZLB era ($+0.64$ – $+0.78$) and weakens substantially in the Hiking era ($+0.19$ – $+0.21$).

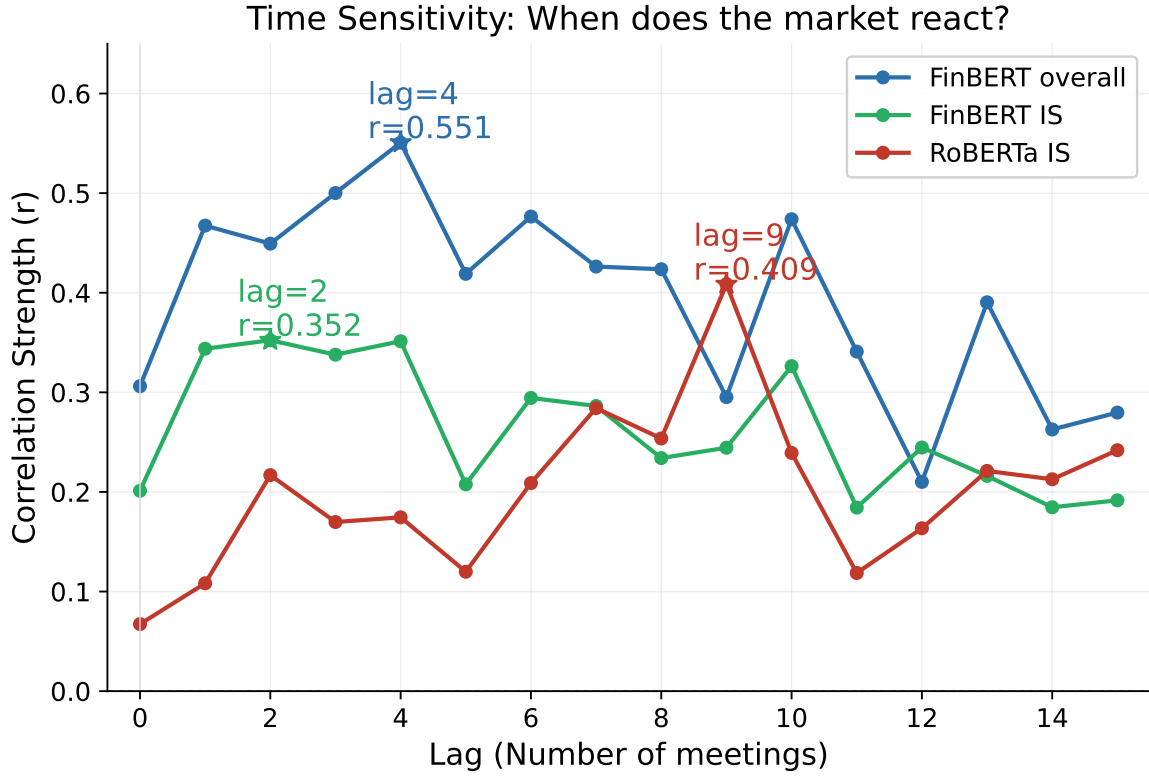


Figure 3.11: Pearson correlation between three sentiment features and MRO changes (changed meetings only, $n \approx 54$).

Level-mean features at lag 4 are more stable across all three regimes.

The feature most consistent across all three eras is **FB** μ $[\mathbf{t}-4]$ (Pre-ZLB: +0.85; ZLB: +0.22; Hiking: +0.61), with **FB** **MP** μ $[\mathbf{t}-4]$ following closely at +0.71 / +0.28 / +0.62. Both are retained for the predictive model on the basis of cross-era robustness. **FB** **IS** $\max[\mathbf{t}-2]$ is also retained: it has the highest full-sample correlation ($r = 0.61$) and a strong pre-ZLB signal (+0.68), providing short-horizon IS information that the mean features do not; its negative ZLB loading and weaker hiking signal are handled by the regime-aware cross-validation.

Table 3.4: Top sentiment features ranked by absolute full-sample correlation. $[\mathbf{t-k}]$: lag in meetings.

Feature	Pre-ZLB	ZLB	Hiking	Full
FB IS max $[\mathbf{t-2}]$	+0.68	-0.18	+0.21	+0.61
RB IS·EP max $[\mathbf{t-2}]$	+0.72	+0.07	-0.31	+0.59
FB max $[\mathbf{t-2}]$	+0.64	-0.18	+0.19	+0.58
RB IS max $[\mathbf{t-2}]$	+0.78	0.00	-0.31	+0.57
FB QA·MP μ $[\mathbf{t-4}]$	+0.55	+0.28	+0.57	+0.56
FB MP μ $[\mathbf{t-4}]$	+0.71	+0.28	+0.62	+0.56
FB μ $[\mathbf{t-4}]$	+0.85	+0.22	+0.61	+0.55
FB QA μ $[\mathbf{t-4}]$	+0.67	+0.24	+0.60	+0.55
RB max $[\mathbf{t-5}]$	+0.69	0.00	+0.12	+0.54
RB EP max $[\mathbf{t-5}]$	+0.66	-0.07	+0.12	+0.53

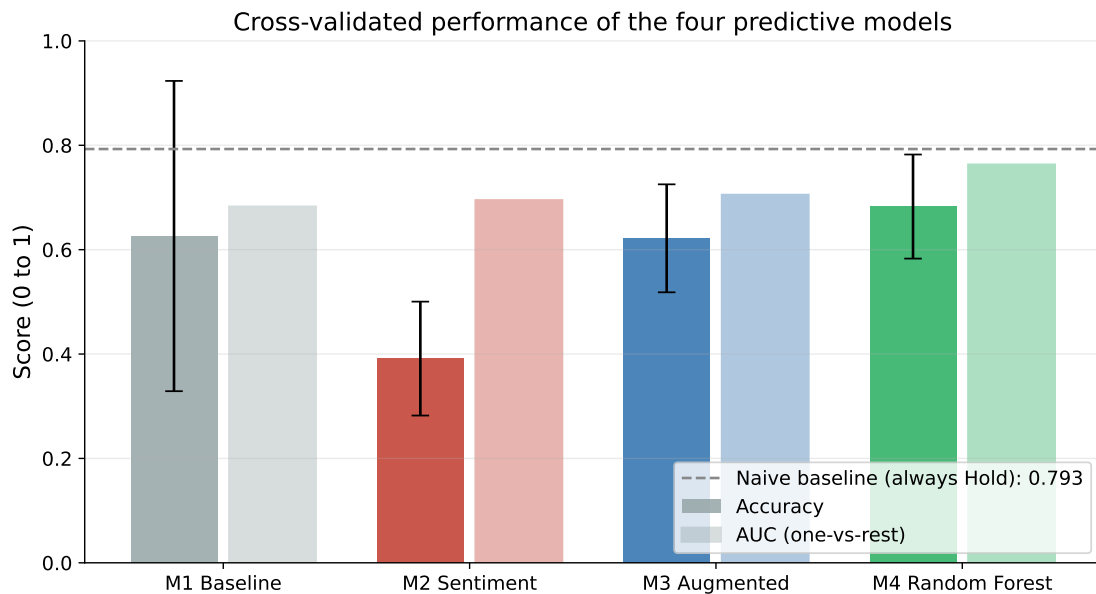
Full: ΔMRO , $n = 54$. ZLB: Wu-Xia shadow rate. Hiking $n = 18$.

3.4.2 Predicting the Direction of ECB Rate Decisions

The predictive task is a three-class classification problem. Each press conference is assigned $y_t \in \{-1, 0, +1\}$ corresponding to the sign of the MRO change: -1 if cut, 0 if held, and $+1$ if hiked. The class distribution is heavily skewed towards *hold* (79.3% of 279 meetings), setting a demanding naive baseline.

Four classifiers are compared. **M1 (Baseline)** uses rate history only: $\{\Delta\text{MRO}_{t-1}, \text{MRO}_{t-1}\}$. **M2 (Sentiment)** uses four features—FinBERT overall mean at lags 2 and 4, FinBERT IS maximum at lag 2, and FinBERT IS standard deviation—isolating the marginal content of the textual signal. **M3 (Augmented)** combines M1 and M2 features. **M4 (Random Forest)** applies a 400-tree ensemble with restricted depth (`max_depth` = 4) to the M3 feature set, relaxing the linearity assumption. All models use class-balanced loss weights and 5-fold time-series cross-validation via *scikit-learn*’s *TimeSeriesSplit*, which constructs five strictly forward-looking, expanding-window train/test splits. The test fold of every split is chronologically posterior to its training fold, so no future information enters the feature set at training time.

Figure 3.12 reports the results. Three observations follow. First, M2’s raw accuracy (0.39 ± 0.11) falls well below the naive baseline—the expected outcome of a class-balanced classifier on an imbalanced sample. With Hold meetings comprising 79.3% of the sample, the always-Hold predictor achieves an accuracy of 0.793 that no class-balanced classifier can mechanically match; we therefore report Area Under the Curve (AUC) (one-vs-rest, macro-averaged) as the primary metric, since it is invariant to the class prior and rewards correct directional ranking rather than majority-class prediction. Second, the AUC

Figure 3.12: Cross-validated performance ($n = 277$ – 279).

ordering is strict: $M1 (0.682) < M2 (0.694) < M3 (0.705) < M4 (0.763)$. Third, and most strikingly, the largest single increment is $M3 \rightarrow M4 (+0.058)$: the Random Forest captures non-linear interactions that the logistic specification cannot, an interpretation confirmed by the importance profile in Figure 3.15.

Figure 3.13 places the M3 and M4 confusion matrices side by side; Table 3.5 reports the underlying precision, recall, and F1 scores.

Table 3.5: Per-class precision (P), recall (R), and F1 for M3 and M4. Full-sample fit; $n = 277$.

Class	M3 Augmented Logit			M4 Random Forest			n
	P	R	F1	P	R	F1	
Cut	0.34	0.73	0.46	0.50	0.83	0.62	30
Hold	0.93	0.61	0.73	0.96	0.79	0.87	219
Hike	0.35	0.86	0.49	0.55	0.93	0.69	28
Macro avg	0.54	0.73	0.56	0.67	0.85	0.73	

The predictive sample loses four hold meetings due to lag-feature construction ($n = 219$)

Both models are strong on the directional classes (Cut and Hike recall 73–93%) but weaker on hold timing: M3 misclassifies 86 of 219 holds as directional decisions. This is the expected behaviour of a signal that leads the rate by several meetings—it correctly anticipates the *direction* of the next move but cannot pinpoint *which* hold meeting precedes it. M4 recovers 40 of those misclassified holds (173 vs 133 correct), accounting

Confusion matrices: M3 Augmented Logit vs M4 Random Forest

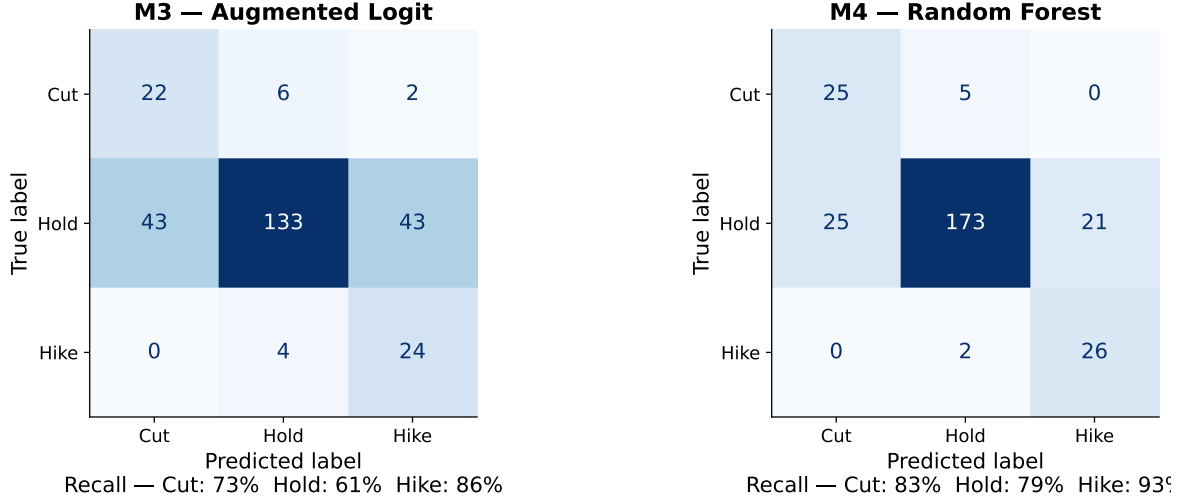


Figure 3.13: Confusion matrices of M3 (Augmented Logit, left) and M4 (Random Forest, right).

for most of the AUC gain.

The coefficient and importance profiles are mutually consistent. In M3, ΔMRO_{t-1} is the sharpest short-horizon predictor: strongly positive on *hike*, strongly negative on *cut*. MRO_{t-1} loads positive on *hold*, capturing the ECB’s low sequential reversal rate. Among sentiment features, FinBERT IS maximum at lag 2 carries the largest hike coefficient (standardised $\approx +1.0$): the most extreme hawkish passage in the IS two meetings prior is the strongest textual precursor of tightening. FinBERT IS standard deviation loads positive on *cut* and near zero on *hold*: a volatile IS signals an upcoming decision, not continued inaction. In M4, FinBERT overall mean at lag 4 is the highest-importance feature (impurity decrease ≈ 0.205), confirming the cross-era robustness identified in Table 3.4; all six features contribute meaningfully, with no single variable dominating. We additionally include FB $\mu[t-2]$ to capture the short-horizon lead identified at lag 2 in 3.11, and FB IS std as a proxy for communication-event volatility; FB MP $\mu[t-4]$ is dropped because of strong collinearity with FB $\mu[t-4]$ ($\rho = 0.81$.)

Extension: composite policy target. Replacing the target with a composite indicator (ΔMRO outside ZLB; Wu–Xia shadow rate [29] within) increases the number of changed meetings to $n_{\text{changed}} = 120$ and shifts the naive baseline to 57.3%. Figure 3.16 shows the result; Table 3.6 reports the classification metrics.

All AUC values fall uniformly relative to the MRO-only setting: M1 (0.576) < M2 (0.657) < M4 (0.668) < M3 (0.699). Two findings stand out. First, M3 remains the best model, confirming that combining rate history with sentiment adds value even on the harder target. Second, M4 no longer outperforms M3: the Random Forest’s

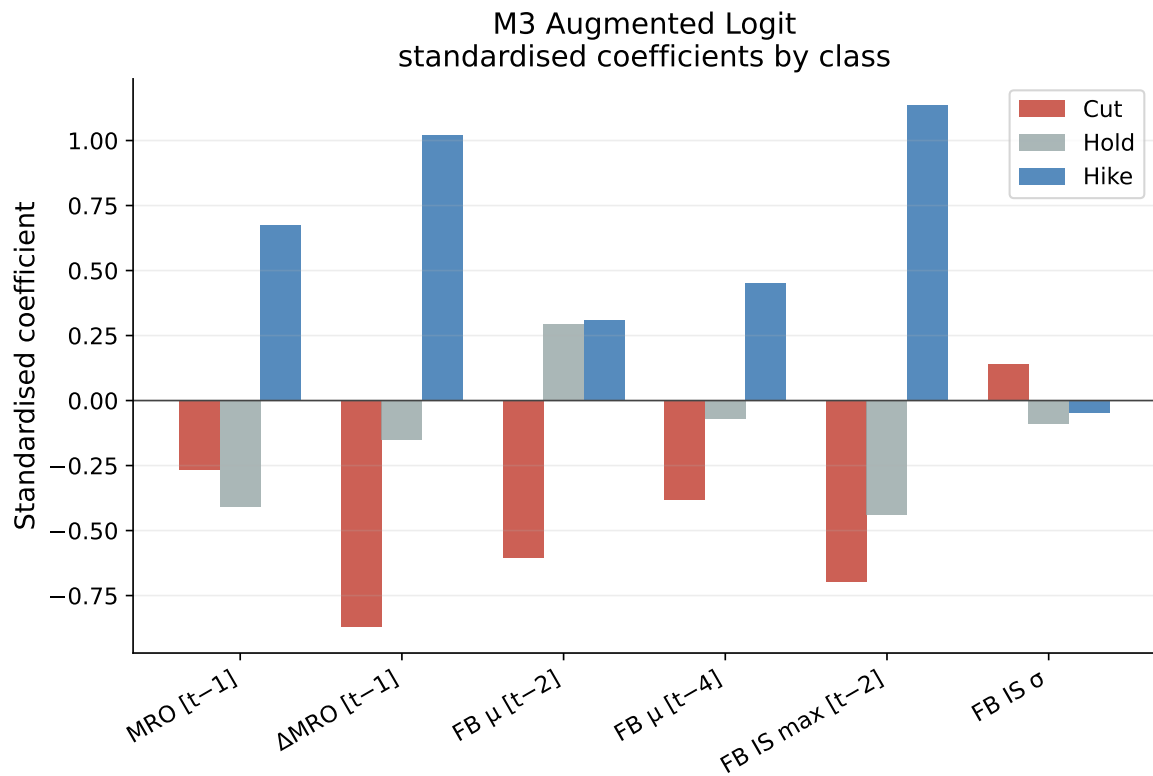


Figure 3.14: Standardised multinomial logistic coefficients of M3 by target class.

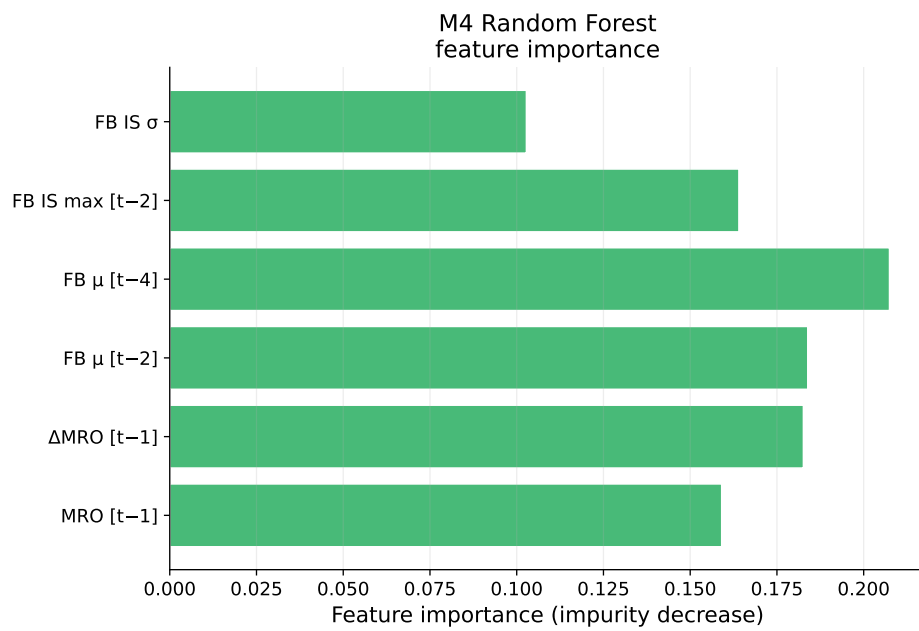


Figure 3.15: Impurity-decrease feature importance of M4.

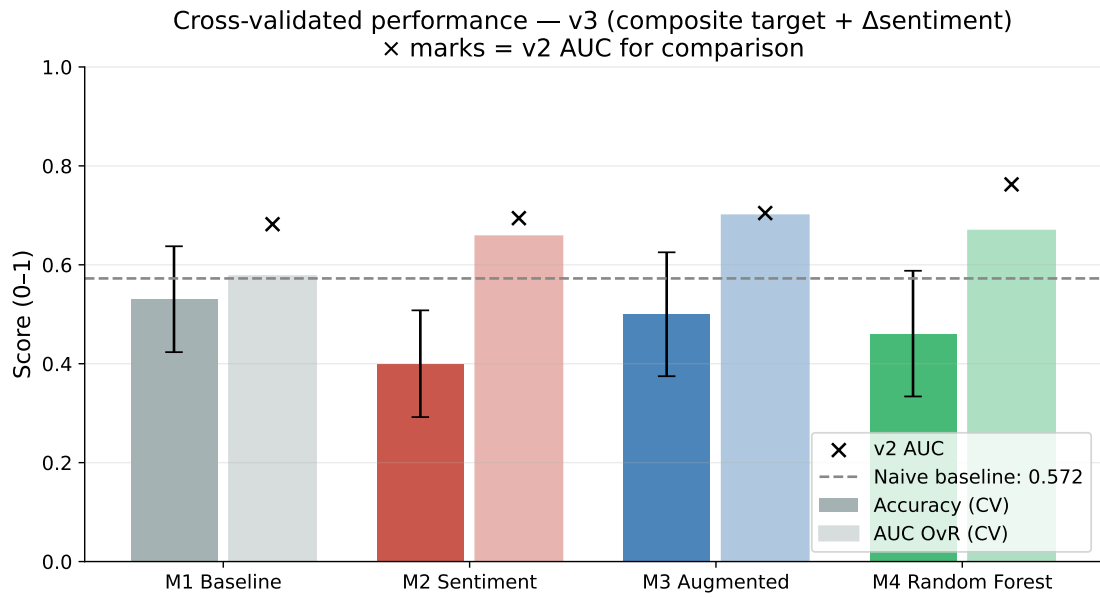


Figure 3.16: Cross-validated performance, composite target ($n = 276$ – 279). ×: v2 AUC for reference. Dashed: naive baseline (0.572).

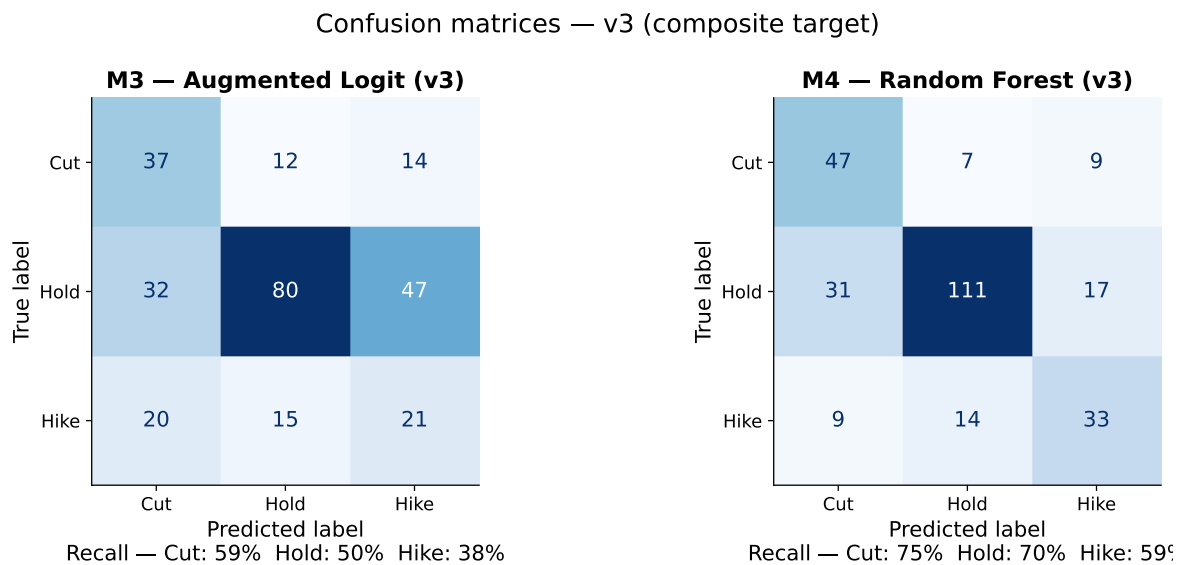


Figure 3.17: M3 (left) and M4 (right) confusion matrices, composite target.

Table 3.6: Full-sample metrics, composite target; $n = 278$.

Class	M3 Augmented Logit			M4 Random Forest			n
	P	R	F1	P	R	F1	
Cut	0.42	0.59	0.49	0.54	0.75	0.63	63
Hold	0.75	0.50	0.60	0.84	0.70	0.76	159
Hike	0.26	0.38	0.30	0.56	0.59	0.57	56
Macro avg	0.47	0.49	0.46	0.65	0.68	0.65	

non-linear advantage is specific to the conventional rate cycle and does not generalise to the composite window, where shadow-rate changes carry different statistical properties. The composite results are best read as a robustness check: directional predictability of sentiment survives the extension to unconventional policy, but the non-linear gain identified in Section 3.4.2 is regime-specific.

3.4.3 Communication Surprise and Market Volatility

The preceding analysis asked whether lagged sentiment predicts future rate decisions. A complementary question is whether the *change* in sentiment from one meeting to the next—the communication surprise—correlates with the contemporaneous market reaction, and whether the *dispersion* of sentiment within a press conference predicts the size of that reaction. Two estimation strategies address these questions: a static full-sample Pearson correlation and a 20-meeting rolling window that reveals regime-level variation. Figure 3.18 reports the static checks.

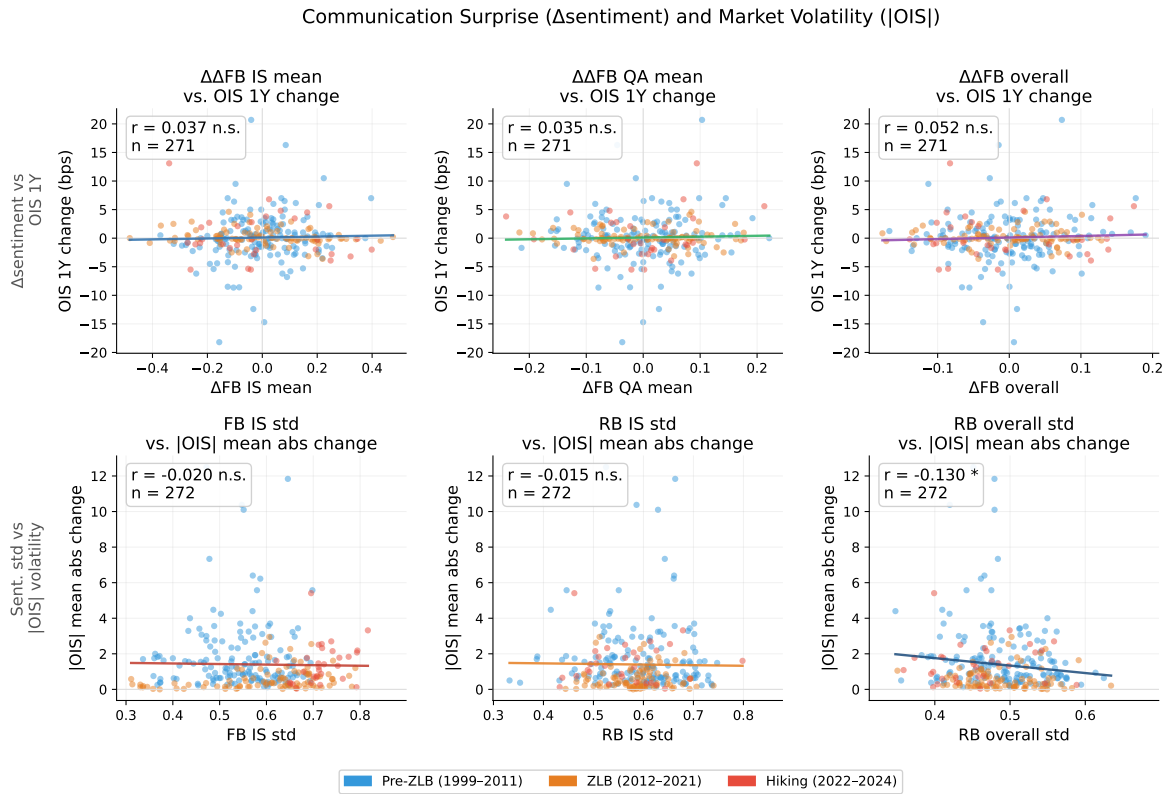


Figure 3.18: **Top row:** Δ FinBERT sentiment (first difference) vs. contemporaneous OIS-1Y change ($n = 271$). **Bottom row:** FinBERT and CentralBankRoBERTa within-conference standard deviation vs. mean absolute OIS change $|\Delta\text{OIS}|$ ($n = 272$). Points colour-coded by era: blue = Pre-ZLB (1999–2011), orange = ZLB (2012–2021), red = Hiking (2022–2024). Regression lines fitted over the full sample.

Static correlations

Three results emerge from the full-sample analysis. First, all directional correlations (top row) are statistically indistinguishable from zero: the first-differenced IS mean ($r = +0.037$, n.s.), Q&A mean ($r = +0.035$, n.s.), and overall FinBERT score ($r = +0.052$, n.s.) carry no systematic information about the sign of the same-day OIS-1Y move. A hawkish communication surprise did not reliably produce a positive rate reaction at the one-year horizon.

Second, the within-conference sentiment dispersion features—proxies for the internal coherence of the press conference—are likewise uninformative about market volatility in the aggregate: FinBERT IS standard deviation ($r = -0.020$, n.s.) and CentralBankRoBERTa IS standard deviation ($r = -0.015$, n.s.) produce near-zero correlations with $|\Delta \overline{\text{OIS}}|$. Third, sentiment *levels*, rather than dispersions, show a weak but statistically significant negative association with market volatility: FinBERT IS mean ($r = -0.146$, $p < 0.05$), FinBERT overall mean ($r = -0.125$, $p < 0.05$), and CentralBankRoBERTa overall standard deviation ($r = -0.130$, $p < 0.05$). The sign is consistent across models: more positive or more coherent ECB communication is associated with *smaller* absolute market reactions, suggesting that communication clarity reduces pricing uncertainty rather than amplifying it. This pattern echoes the communicative-cancellation finding of Section 3.2.3.

Time-varying structure

The static aggregate conceals a strongly regime-dependent structure. Figure 3.19 traces the same directional and volatility correlations through a 20-meeting rolling window (≈ 2.5 years at eight meetings per year). The dotted horizontal lines mark the two-sided 95% significance threshold for $n = 20$ ($r_{\text{crit}} = \pm 0.444$); segments outside the corridor are statistically distinguishable from zero at the local-window level, while segments inside should be read as noise.

The top panel reveals three structurally distinct regimes. In the pre-GFC period the IS surprise correlation peaks at approximately $+0.50$, marginally clearing the significance corridor: a hawkish tone shift moved one-year rates upward on impact, consistent with an environment in which markets treated ECB communication as a reliable signal of the near-term policy path. This relationship collapsed during the GFC and turned persistently and significantly negative through the ZLB era, reaching a trough of approximately -0.50 to -0.60 during 2010–2013: under forward guidance and the effective lower bound, a hawkish communication surprise was re-interpreted as a reassurance about the economic outlook rather than a tightening signal, producing the opposite market response. The hiking cycle (2022–2024) shows a tentative recovery toward positive values, though the rolling correlation remains inside the significance

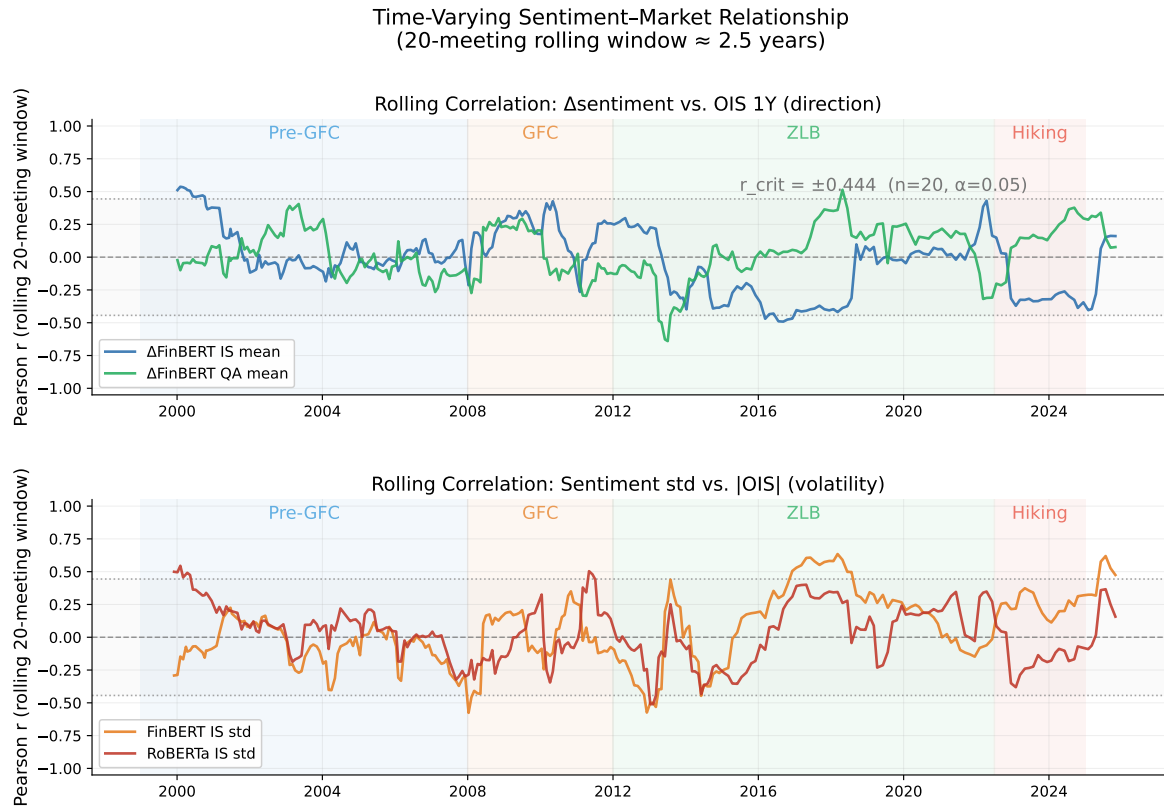


Figure 3.19: Rolling Pearson correlation over a 20-meeting window (≈ 2.5 years). **Top:** Δ FinBERT sentiment vs. OIS-1Y change (direction). **Bottom:** FinBERT and CentralBankRoBERTa IS standard deviation vs. $|\Delta \text{OIS}|$ (volatility). Dotted horizontal lines: two-sided $\alpha = 0.05$ significance threshold for $n = 20$ ($r_{\text{crit}} = \pm 0.444$); segments outside the corridor are significantly different from zero. Window length is chosen as the shortest interval that remains within a single monetary-policy regime; see text.

corridor for most of that period, indicating that the sign reversal had not yet fully stabilised by the end of the sample. Crucially, the near-zero full-sample correlation reported above is an artefact of temporal aggregation: pre-GFC positive and ZLB-era negative effects largely cancel in the pooled estimate.

The bottom panel tells a more muted story. The within-conference sentiment dispersion measures spend the majority of the sample inside the r_{crit} corridor, consistent with their full-sample non-significance. Two brief exceptions are visible: a spike toward +0.50 during the GFC (2008–2009), when elevated market volatility and elevated internal communication uncertainty coincided, and a cluster of positive values during mid-ZLB crisis episodes (2011–2013). In the hiking cycle the dispersion–volatility correlation collapses toward zero or reverses, a pattern that is economically interpretable: the ECB communicated with unusual resolve during 2022–2023 (low internal dispersion) while simultaneously producing the largest OIS moves in the sample (high volatility), mechanically suppressing the correlation. The regime instability is itself an economically meaningful finding: the informational content of communication surprises, and of internal coherence signals, depends critically on the prevailing monetary policy framework rather than on any stable structural relationship between language and prices.

Summary of Chapter Findings. Across the four arcs of analysis, four substantive findings have emerged. First, the two sentiment classifiers produce coherent, mutually validating signals that align with the major historical turning points of euro-area monetary policy. Second, the prepared IS and the spontaneous Q&A are structurally different communicative acts, and the journalist–ECB gap is a stable institutional feature that, if not separated by the pipeline, would have contaminated the policy signal by approximately 0.3 on the unit scale. Third, sentiment leads the ECB’s rate decisions by between two and ten meetings, with the lead being long and gradual in the pre-ZLB era and approximately symmetric in the post-ZLB hiking cycle. Fourth, sentiment features carry incremental predictive content beyond the rate history itself: AUC rises from 0.682 (M1, rate-history only) to 0.694 (M2, sentiment only) and 0.705 (M3, augmented logit); a Random Forest on the same features adds a substantial further increment to 0.763, indicating genuine non-linear interactions that the logistic specification cannot capture.

Chapter 4

Discussion

The empirical findings of Chapter 3 establish that the textual signal carried by the ECB’s press conferences is internally coherent, structurally heterogeneous between its prepared and spontaneous components, leading rather than coincident with the policy rate, and weakly but systematically informative about the direction of the next rate decision. This chapter discusses what these findings imply for policy makers and market participants (Section 4.1), what they reveal about the methodological choices that shaped them (Section 4.2), the limitations that bound their interpretation (Section 4.3), and the most natural extensions of the framework (Section 4.4).

4.1 Implications for Policy Makers and Markets

The headline empirical regularity of the thesis is that affective tone leads the policy rate rather than tracking it coincidentally. The event study of Section 3.3.3 shows a pre-ZLB pattern in which a hawkish profile is fully established up to six meetings before an actual hike, and a post-ZLB pattern in which both hikes and cuts are symmetrically telegraphed across the same window. The lag-correlation analysis of Section 3.4.1 corroborates this in a non-event-study setting and identifies a peak correlation between the FinBERT overall mean and the MRO change at a lag of four meetings ($r = 0.551$, changed meetings only, $n \approx 54$). From the perspective of a market participant, this means that the textual record of ECB communications is a leading indicator of the price of money rather than a contemporaneous one: the headline-only consumer of ECB decisions is processing information that has been embedded in language across the preceding policy cycle. From the perspective of a policy maker, it means that the post-pandemic ECB’s strategy of long, symmetric telegraphing of both directions of policy change is reflected in the price formation of euro-area rate expectations to a measurable degree—a finding consistent with the broader literature on the effectiveness of central bank forward guidance [4, 28].

A second implication concerns the structural division of the press conference. Section 3.2.2 demonstrated that the prepared IS has a coherent three-act narrative arc in which the most market-relevant policy content is concentrated at the end—a swing of approximately 0.4 sentiment units in FinBERT from the risk-catalogue trough to the policy-decision close—while the Q&A has no comparable arc and is best characterised as a flat reactive dialogue. A market that has learned this structure should already be paying disproportionate attention to the closing paragraphs of the IS. The implication for market participants is the symmetric one: the affective tone of the Q&A, while not without informational value, should be read as a divergence signal relative to the prepared text rather than as a primary signal in its own right. This structural decomposition complements the textual analysis of Pavelková [24], who documents the narrative anatomy of ECB press conference transcripts, and the broader evidence that ECB verbal communication moves financial markets [17].

A third implication, and perhaps the most subtle, concerns the communicative-cancellation pattern of Section 3.2.3. The directional signal in the data is that press conferences in which IS and Q&A sentiment diverge most tend to produce the *smallest* OIS reactions ($r = -0.146$, $p_{\text{raw}} = 0.016$); however, this association does not survive Holm–Bonferroni correction for the family of six simultaneous tests ($p_{\text{adj}} = 0.096$) and therefore cannot be treated as a confirmed finding. Its direction is nonetheless theoretically interpretable: the market plausibly reads internal disagreement between the prepared narrative and the contested interpretation surfaced during questioning as evidence that no clean policy update is on offer. This reading is consistent with the view of the press conference as a coordination device, in which coordination failures reduce informational content rather than increase it. The result should be revisited with additional data before any strong claim is made.

A fourth implication concerns the structural pessimism of the journalist corps. Section 3.2.3 documented that journalist questions are systematically more dovish than the ECB’s answers by an average of 0.29 on the unit scale, and that this gap is stable across structurally different monetary regimes. The institutional reading is clear: the press corps is professionally rewarded for surfacing downside risks, while the central bank is constitutionally bound to project measured stability; the gap is a structural feature of the discourse rather than a measurement error. The practical implication for any future sentiment-based research on central bank communication is that the question–answer separation is not an optional refinement but a methodological prerequisite: without it, the affective tone of the press corps contaminates the policy signal by an order of magnitude that is large relative to the policy-relevant variation in the signal itself.

Finally, the predictive content of sentiment for the direction of rate decisions carries implications of its own. The strict AUC ordering $M1 (0.682) < M2 (0.694) < M3 (0.705)$

$< M4$ (0.763) (Section 3.4.2) establishes two claims of independent interest. First, the M2-versus-M1 contrast shows that the textual record adds directional information beyond what is contained in the rate history alone. Second, the M4-versus-M3 gap (+0.058) indicates that part of this informational content is carried by non-linear interactions that the logistic specification cannot capture. Both claims are consistent with the conclusion that ECB verbal communications are not redundant with their policy actions: the text adds signal.

4.2 Methodological Reflections

The empirical results of Chapter 3 cannot be separated from the methodological choices that produced them. Four choices deserve explicit reflection.

Sentiment versus stance. The framework adopted in this thesis measures the *affective tone* of ECB discourse rather than its *policy stance*. The distinction matters. Sentiment classifiers such as FinBERT [2] and CentralBankRoBERTa [25] assign a positive/negative polarity to a passage on the basis of its emotional valence; stance classifiers assign a hawkish/dovish/neutral label on the basis of its policy implication. The two need not coincide. A sentence such as “*inflation is far too high and we are determined to bring it back to target*” carries a strongly negative emotional valence (alarm about inflation) but implies a hawkish policy stance (commitment to tighter policy). The ZLB-era flatness of the sentiment series (Section 3.3.1) is a direct manifestation of this mismatch: the language of unconventional easing—“asset purchase programme”, “reinvestment”, “favourable financing conditions”—is lexically distinct from the language of conventional rate cuts, and a classifier whose training distribution is dominated by the latter will not register the former as dovish [30]. The state-of-the-art stance classifiers fine-tuned specifically on FED and ECB corpora [26, 27] would in principle close this gap, but they are released under gated access on the Hugging Face Hub and were not obtainable within the timeframe of this study. The signal extracted here should therefore be read as affective tone, and the ZLB flatness as a property of the training distribution of the classifiers rather than of the ECB’s communicative behaviour.

Granularity and the chunk-size trade-off. Section 2.2.2 reported that the Pearson correlation between the FinBERT and CentralBankRoBERTa meeting-level means declines monotonically from $r = 0.727$ at single-sentence granularity to $r = 0.589$ at 350 words, while the within-meeting standard deviation stabilises beyond 200 words ($\Delta\bar{\sigma} < 0.001$ thereafter). The trade-off is genuine. Short chunks are evaluated more consistently by two independent encoders because their context window is shorter and the encoders rely on lexical cues that they share; the same shortness, however, fragments

coherent economic arguments into decontextualised pieces and erodes the contextual reasoning that motivates transformer encoders in the first place. The 200-word setting adopted as the headline configuration trades stability for context; the single-sentence configuration used in the predictive analysis of Section 3.4.2 trades context for predictive power on a small sample. Neither configuration is uniformly superior, and the choice is best understood as a function of the empirical question rather than as a methodological invariant.

Topic taxonomy. Section 2.2.2 reported that the initial ten-topic taxonomy resulted in decision paralysis for the zero-shot classifier and was therefore simplified to four core themes with descriptive hypothesis strings, following the NLI benchmarking approach of Yin et al. [31]. This simplification is a pragmatic compromise rather than a principled choice. A four-topic taxonomy is too coarse to distinguish, for example, between forward guidance and inflation analysis within the broader Monetary Policy category, and the resulting topic-conditioned sentiment series are correspondingly less informative than they could be at a finer granularity. The robustness check with a second NLI backbone (DeBERTa-v3 [21] trained on a broader collection of NLI corpora) attenuates this concern by demonstrating that the classifier output is not an artefact of the specific encoder, but it does not resolve the underlying limitation that the four-topic taxonomy reflects what zero-shot classification can do reliably rather than what the empirical question would ideally require.

Linear baselines for a small-sample problem. The predictive analysis of Section 3.4.2 relies on a multinomial logistic regression as the primary specification and a random forest as a non-linear robustness check. The choice of a linear baseline reflects the small sample of changed meetings ($n \approx 56$ over 25 years) rather than a methodological preference. With approximately 56 effective observations and the four-feature set used in models M2 and M3, the ratio of observations to parameters is at the edge of the range in which logistic regression remains identifiable, and richer architectures (gradient-boosted trees, neural networks) would face an acute over-fitting risk. The random forest M4 was included precisely because it is the simplest non-linear specification with a built-in mechanism for handling small samples (bootstrap aggregation, restricted depth) and therefore provides the most defensible test of whether the linear specification is leaving signal on the table. The marginal AUC gain of M4 over M3 (+0.058) is the largest single increment in the model comparison and indicates that part of the predictive content is carried by non-linear interactions; the moderate absolute level of M4 (0.763) is consistent with the interpretation that the sample size, not the specification, is the binding constraint.

4.3 Limitations

Four limitations bound the interpretation of the results above.

Sample size. The sample consists of 272 press conferences over the 1999–2024 window. Of these, only approximately 56 involve a change in the MRO rate, with the remaining meetings being either holds or covered by the ZLB flatness. This structurally small number of decision events—driven by the fact that the ECB holds eight meetings a year and changes rates infrequently—limits the overall statistical power of the analysis. While the 5-fold time-series cross-validation mitigates this concern relative to a single train–test split, each fold inherently contains a very small number of changed meetings. As a result, the AUC estimates carry a non-trivial across-fold standard deviation that should always be read alongside the point estimates.

Restriction to a single communication channel. The analysis is restricted exclusively to ECB press conference transcripts. While press conferences represent the most high-profile communication window, the ECB generates a vast ecosystem of alternative text types—including the Economic Bulletin, formal Monetary Policy Accounts, and hundreds of public speeches by Governing Council members. The findings of this thesis should therefore be read as statements about the immediate communication dynamics of the press conference window, rather than a comprehensive assessment of the ECB’s total communication footprint. The framework, however, is portable: the scraping, preprocessing, and classification pipeline documented in Chapter 2 could be adapted to these alternative ECB textual channels without structural methodological changes.

Affective scope and ZLB mismatch. As discussed in Section 4.2, the framework captures affective sentiment rather than policy stance. This limitation manifests in two ways. First, the binary positive/negative polarity discards the neutral class, meaning that descriptive or procedural language is mapped to zero. This attenuates the aggregate signal, particularly during routine press conferences. Second, the classifiers lack the domain-specific vocabulary of unconventional monetary policy. Their flatness during the ZLB window is a documented limitation of their training distribution rather than a true reflection of the ECB’s communicative intent, meaning the predictive outputs during the ZLB period must be interpreted with caution.

4.4 Future Work and Groundwork for Publication

The framework of this thesis is designed to be extended. Four extensions are particularly natural.

Stance classification. The single most informative extension would replace the affective-sentiment layer with a dedicated stance classifier. The gated classifiers of Shah et al. [26, 27] represent the current state of the art, but their restricted access on the Hugging Face Hub could not be obtained within this study’s timeframe, and no collaboration with the authors was pursued. Two independent paths remain open. First, a custom stance classifier can be trained from scratch on a manually annotated subset of the existing ECB chunk corpus using the annotation protocol of [26]; a corpus of approximately 2,000 labelled chunks would be sufficient for fine-tuning a RoBERTa-base encoder and would require only a modest annotation effort. Second, a proprietary Large Language Model (e.g. GPT-4o or Claude) can serve as a zero-shot stance annotator with a carefully designed prompt, providing a high-quality silver-standard label set for distillation into a smaller, deployable encoder. This second path is the most modern approach currently available, though it introduces a critical caveat: Large Language Model (LLM) outputs are *non-deterministic* unless temperature is fixed at zero, meaning the same chunk can receive different labels across runs; any pipeline that uses an LLM as a labeller must therefore fix temperature and log the exact model version to ensure reproducibility.

Fine-tuning the topic and stance classifiers. The zero-shot topic classifier is limited by the expressiveness of its four hypothesis strings and the general-purpose NLI pre-training of its backbone. A labelled corpus of ECB chunks would unlock two improvements. First, the topic model can be fine-tuned directly: starting from `facebook/bart-large-mnli` or `DeBERTa-v3` and continuing training on a domain-specific NLI dataset (premise = ECB chunk; hypothesis = topic string; label = entail/neutral/contradict) would substantially narrow the classifier’s uncertainty on ambiguous passages. Second, a fine-tuned stance classifier—trained on the same corpus with hawkish/dovish/neutral labels—would subsume both tasks, providing a single model that jointly assigns policy intent and thematic focus. The modular database architecture of Section 2.3 supports both extensions without schema changes: new `topic_model` rows and a new `version` on `topic_labels` accommodate additional label sets, and the `model_id` foreign key on `topics` allows results from old and new classifiers to coexist in the same store for direct comparison.

Intraday market windows. The market panel used in this thesis aggregates the OIS reaction at the daily level. The Euro Area Monetary Policy Event-Study Database (EA-MPD) [1] also provides intraday windows around the press release (14:15 CET) and around the press conference (14:45–15:15 CET). Separating these windows would allow the predictive analysis to test whether the affective tone of the IS is incorporated into rates faster than the affective tone of the Q&A. The hypothesis that the Q&A signal is the slower-incorporating one is consistent with the divergence finding of Section 3.2.3 and is directly testable with the existing data.

Multi-channel ECB textual expansion and NBS publication groundwork.

The pipeline is portable across distinct text sources. A natural follow-up study would extend the analysis to the full spectrum of ECB verbal outputs, specifically incorporating executive speeches and the Monetary Policy Accounts under identical chunking, classification, and sentiment extraction rules. This would allow a structured comparison of the lead–lag profiles and sentiment trajectories across different communication channels within the same institution. From the perspective of the planned co-authored publication with the Národná banka Slovenska, this intra-ECB expansion is the most direct policy-relevant extension: mapping the communication asymmetry between formal statements and decentralized speeches would directly inform the conditions under which a small open economy should treat different ECB channels as leading indicators versus contemporaneous noise. The codebase developed for this thesis—the scraping pipeline of Section 2.1, the database schema of Section 2.3, the chunk-to-meeting aggregation operator of Section 2.5.2, and the predictive framework of Section 3.4—is structured as a configurable framework precisely to support this multi-channel text integration.

Concluding remark. The four findings of Chapter 3—internal coherence of the two sentiment series, structural heterogeneity between IS and Q&A, lead–lag asymmetry of sentiment around rate decisions, and incremental predictive value of sentiment beyond the rate history—together establish that the textual record of ECB press conferences carries economically meaningful information beyond what is contained in the policy rate itself. The methodological framework that produced these findings is portable and extensible, and the limitations identified above describe extensions of scope rather than barriers to the central conclusion.

Conclusion

This thesis has examined how machine-learning analysis of the textual record of ECB press conferences quantifies, structures, and ultimately prices the information that the central bank communicates to financial markets. Working with a custom corpus of 289 press conferences from June 1998 to October 2024, and a pipeline that separates the prepared IS from the spontaneous Q&A and, within the Q&A, journalists’ questions from the President’s answers, the empirical chapters delivered four findings.

First, FinBERT and CentralBankRoBERTa produce internally coherent series whose turning points align with the canonical episodes of the euro-area cycle: the October 2008 rate cut, Draghi’s 2012 “Whatever it takes” speech, the 2015 QE launch, the 2020 pandemic emergency, and the 2022 hiking lift-off. Two encoder families trained on different corpora agree on the sign of the signal, which is the minimum bar for treating the textual measure as more than a model artefact.

Second, the IS and the Q&A are structurally different communicative acts. The IS traces a coherent three-act narrative arc—moderate opening, risk-catalogue trough, hawkish policy-decision close—with a within-conference swing of roughly 0.4 FinBERT units, while the Q&A is essentially flat as a function of relative position. Within the Q&A, journalist questions are systematically more dovish than the ECB’s answers, by an average of 0.29 on the CentralBankRoBERTa scale, and this gap is stable across the Pre-ZLB, ZLB, and hiking regimes. Without separating the two voices, the press corps’ affective tone would contaminate the policy signal by approximately 0.3 units—an order of magnitude that is large relative to the policy-relevant variation. The structural decomposition of the press conference is therefore not an optional refinement but a methodological prerequisite for any sentiment-based study of central-bank communication, and is one of the principal departures of this thesis from earlier studies that treat the press conference as a single document.

Third, sentiment leads the ECB’s policy decisions rather than tracking them coincidentally. The event-study window of ± 6 meetings shows a hawkish profile fully established up to six meetings before an actual hike in the pre-ZLB period and a symmetric telegraphing of both hikes and cuts in the post-pandemic cycle. The lag-correlation analysis identifies a peak between the FinBERT overall mean and the MRO change at a lag of four meetings ($r = 0.551$ on changed meetings).

Fourth, the textual signal carries incremental predictive content for the direction of the next ECB rate decision. Four classifiers were compared on a three-class target: a rate-history baseline (M1), a sentiment-only logit (M2), an augmented logit (M3), and a Random Forest on the same feature set (M4). The cross-validated AUC ordering is strict: $0.682 < 0.694 < 0.705 < 0.763$. The M2-versus-M1 increment establishes that the text adds directional information beyond the rate history alone; the M4-versus-M3 increment of $+0.058$ shows that an economically meaningful portion of that information is carried by non-linear interactions which the logistic specification cannot capture. The textual record and the rate history are complementary rather than redundant.

Four extensions are particularly natural. First, the affective-sentiment layer should be replaced with a dedicated *stance classifier* that assigns the operationally meaningful hawkish/dovish/neutral label rather than positive/negative tone; the ZLB-era flatness of the present series is a direct artefact of the sentiment–stance mismatch, and a stance-aware classifier would close it. The gated state-of-the-art models of Shah et al. [26, 27] are the most direct route, but a custom classifier fine-tuned on a manually annotated ECB corpus is feasible at modest annotation cost. Second, large language models such as GPT-4o or Claude can serve as zero-shot stance annotators at near-human quality, providing a silver-standard label set for distillation into a smaller deployable encoder; reproducibility requires fixing temperature and logging the model version. Third, the daily OIS reaction used here can be replaced by the intraday windows of the Euro Area Monetary Policy Event-Study Database [1], separating the 14:15 CET press-release window from the 14:45–15:15 CET press-conference window; this directly tests the hypothesis that the Q&A signal is the slower-incorporating of the two. Fourth, and most importantly from the policy perspective, the pipeline can be replicated on the press conferences of the Federal Reserve and the Bank of England under identical chunking, classification, and aggregation rules. This cross-bank replication is the planned next step in the collaboration with the Národná banka Slovenska and forms the empirical core of the envisaged co-authored publication: the cross-bank profile of communication asymmetry will directly inform the conditions under which a small open economy facing the euro area should treat ECB sentiment as a leading indicator versus a contemporaneous signal.

Finally, a brief note on scope is warranted. While the title of this thesis invokes the trading value of central bank information, the core empirical contribution of this work is establishing the predictive value of ECB sentiment for the direction of the next rate decision. The translation of this predictive signal into an actionable, backtested tradable strategy—net of transaction costs and market friction—represents the most natural financial extension of this framework and is identified as a priority for future work.

The qualitative record of monetary policy can be read systematically, separated

cleanly into its prepared and spontaneous components, and converted into a structured time series with measurable predictive value for the price of money. The framework developed in this thesis is one step in that programme; the limitations identified along the way describe extensions of scope rather than barriers to the central conclusion.

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Appendix A: Contents of the Electronic Appendix

The electronic appendix is officially submitted as a compressed archive (`.zip`) through the university information system (AIS2) and contains the complete source code and materials used in this thesis.

The archive is structured into two main directories:

- `source/` – The core Python codebase, including modules for web scraping, LLM-based NLP processing, data integration, predictive modeling, and statistical analysis.
- `praca/` – The $\text{\LaTeX}$ source files, bibliography database, and figures used to compile this document.

For detailed information regarding environment configuration (`requirements.txt`) and step-by-step replication instructions, please refer to the `README.md` file in the root directory.

The latest version of the source code is also publicly available on GitHub at: <https://github.com/kubovotny/bachelor-thesis>.